

Jing-Zhang AI Innovation Belt · Agent Urban-Design Open Call

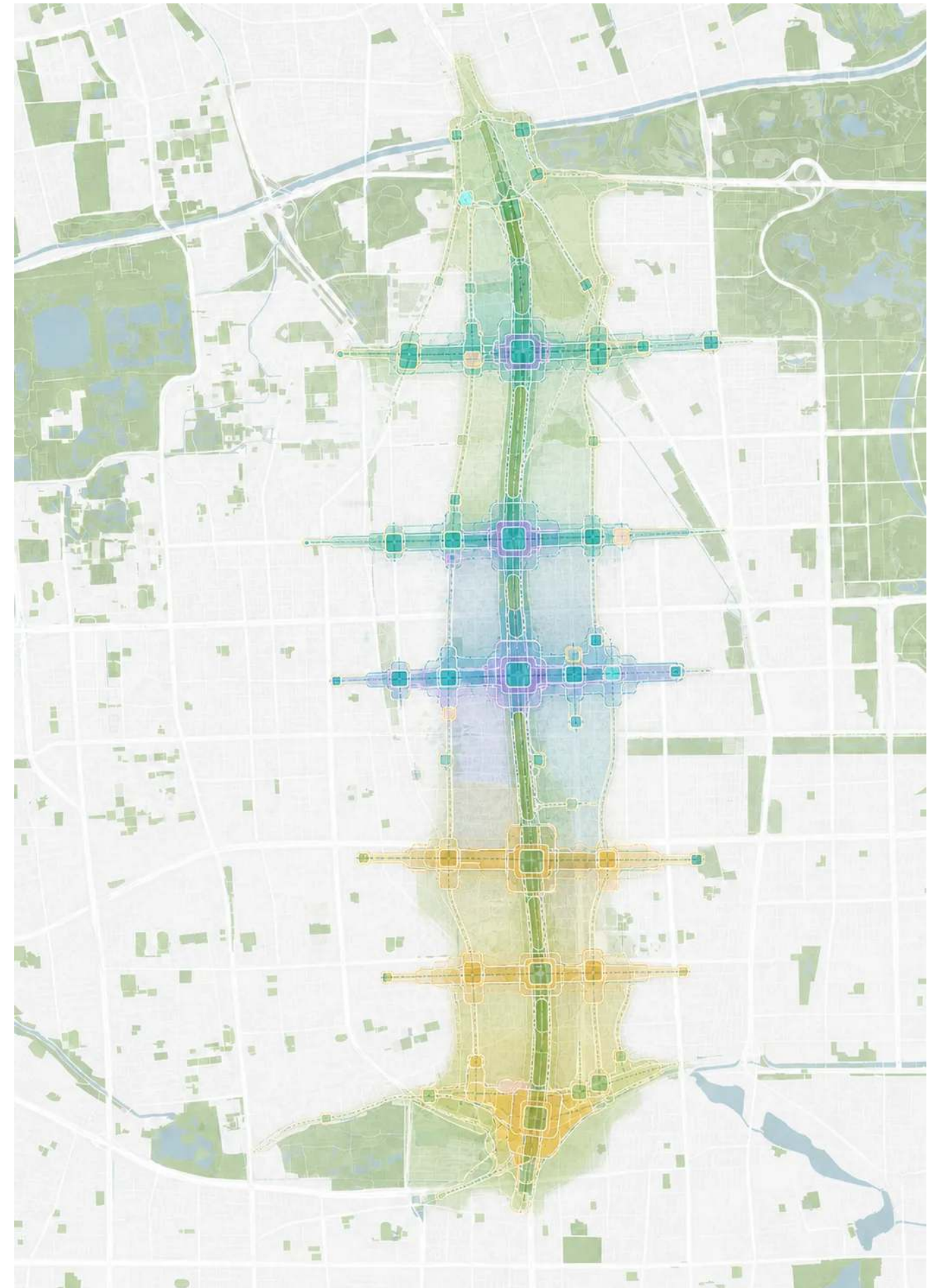
Open-Source Ground: Public Transformation of Knowledge-Production Space along the Jing-Zhang Railway

开源地表：京张知识生产空间的公共转型

A continuous, learnable and reprogrammable public ground links the Jing-Zhang Railway Heritage Park with public ground floors, institutional forecourts, transverse streets, transit, river space and community entrances.

Design team / Agent: Codex x Keldos-Li / Open-Source Ground

Tracks: AI Traffic & Walkability · Enterprise Services Ecosystem · Civic Agent Governance

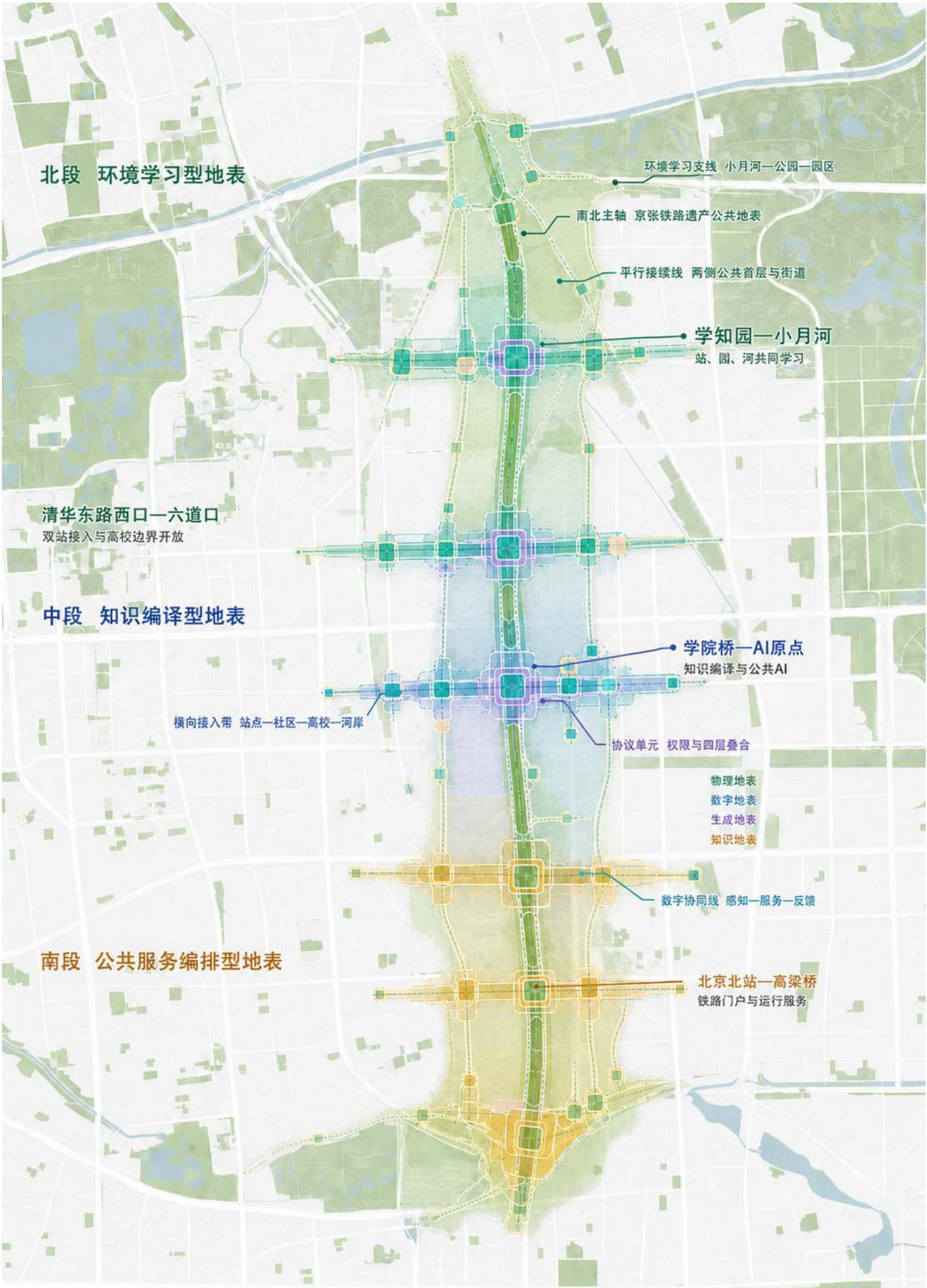


Open-Source Ground

The Jing-Zhang Railway once carried knowledge, people and industry into Beijing. Today its corridor concentrates universities, research institutes, AI companies, transit stations, established communities and urban parks. This density already supports a world-class innovation corridor, yet walls, inaccessible ground floors, transverse breaks and fragmented services divide it into separate internal systems. Open-Source Ground takes the Jing-Zhang Railway Heritage Park as a continuous public base and weaves adjacent ground floors, institutional forecourts, transverse streets, stations, riverbanks and community entrances into an accessible, continuously learning and reprogrammable urban system. Universities, companies, residents and AI systems can then produce knowledge, space and public service together.

A continuous north-south public ground is stitched to campuses, parks, communities and transit through five transverse accesses. Fifteen-Minute Open Units distribute everyday services, while Ground Protocol Units govern entrances, opening hours, permissions, maintenance and recovery. Physical, digital, generative and knowledge grounds share the same plan, parcel and ground floor. The northern section becomes environmental-learning ground, the central section knowledge-compilation ground, and the southern section public-service-orchestration ground. Zhongzhiyuan, Beijing AI Origin and Dazhongsi each receive an independent plan, key section and representative public-life scene.

The design makes “open source” tangible through six urban rights: to see what is open, reach a public interface, use space and facilities, contribute a problem or knowledge, participate in rule revision, and exit a data or technical system. Space, knowledge, data and models, facilities, urban problems, public services and governance interfaces all become resources that people can access.



1 Design Basis and Material Register

The proposal integrates three consecutive bodies of work. Site research establishes spatial facts and problems; comparative case study identifies transferable transformation mechanisms; the final design turns those findings into an overall structure, key–area plans, sections, scenes, projects and operating rules. The two internal studies are navigation layers within the design process. Every formal reference below traces back to an original official, public–institution, academic or open–data source.

The public qualification announcement and Agent Taskbook define a Coordinated Research Area of about 43.6 km², an Overall Design Area of about 11.4 km² and three Key–Area Detailed Design Areas. Site analysis uses an OpenStreetMap Beijing extract retrieved on 12 August 2026, EPSG:4548, a 250 m output grid and a 400 m activity observation window. Twenty–two maps examine location, transit, walking and cycling, barriers, urban fabric, vitality, innovation and education, public service, culture, commerce, blue–green space, renewal clues and key–area differences. [Source](#) [Source](#) [Source](#)

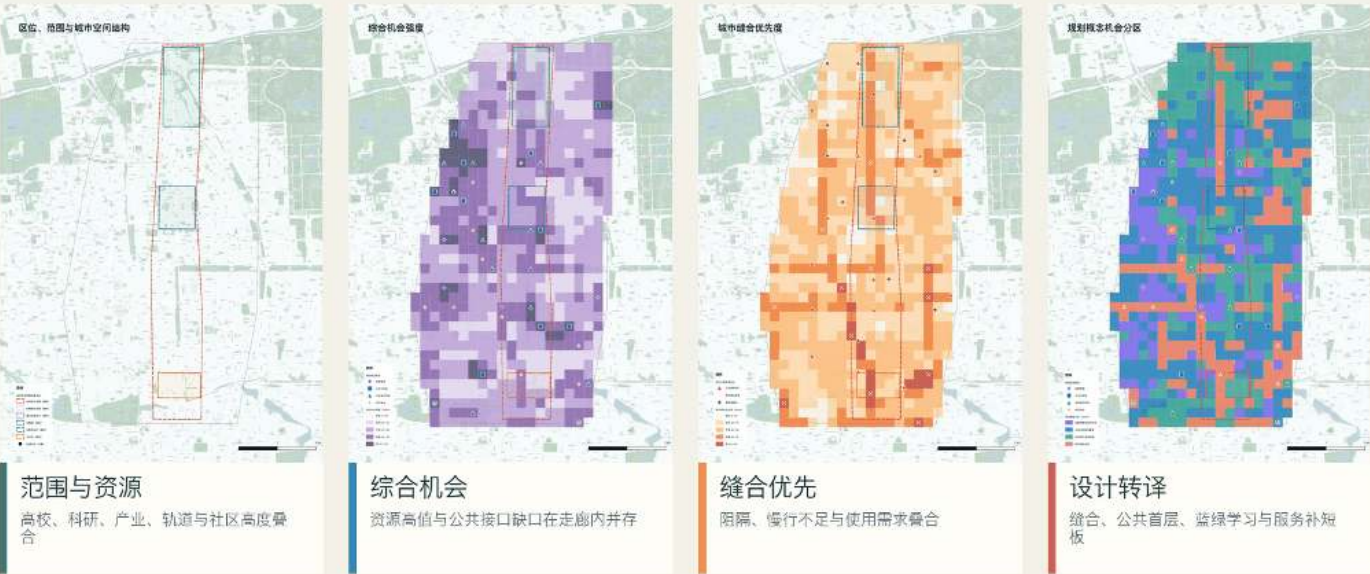
The dataset supports corridor–scale comparison and priority selection. The analysis grid covers about 43.25 km² and identifies 3,975 points of interest. Estimated green space is 13.2%, water 0.7%, and building coverage 17.8%; median straight–line distance to a transit station is about 606 m. Walking and cycling catchments are adjusted for network supply and road/rail barriers. Real entrances, crossing delay, slope, opening hours and lived experience require field calibration.

Twelve in–depth cases were selected from 27 candidates. They address continuous public space, rail stitching, public interfaces of knowledge institutions, blue–green testing, heritage reuse, phased delivery and multi–actor operation. King’s Cross, Paris Rive Gauche and the southern Yangpu Riverfront form the core set. KIC, 22@Barcelona, Kendall Square, Gyeongui Line Forest Park, The Bentway, Stockholm Royal Seaport, Shougang Park, Guiwan Park and Hong Kong Science Park add specialist mechanisms. Every reproduced image is tied to an original page and explicit reuse status.

The current design uses provisional extents. Official key–area boundaries, station exits, road red lines, ownership, building condition, river and flood controls, municipal capacity and rail–crossing conditions remain unknown. Dazhongsi is therefore represented only as a four–direction potential access framework.

01 从场地证据到设计任务

22张专题分析共同识别资源集聚、公共接口稀薄与横向断裂



场地认识 → 问题发现 → 核心矛盾 → 空间动作

证据边界：OSM/BBBike公开数据支持走廊尺度比较；真实入口、开放时段和工程条件需专项核验。

Site–research evidence chain

The evidence chain moves from site reading to problem discovery, core contradiction and spatial action. The twenty–two analytical maps below provide the detailed evidence.

Three-Scale Working Framework

2 Three-Scale Working Framework

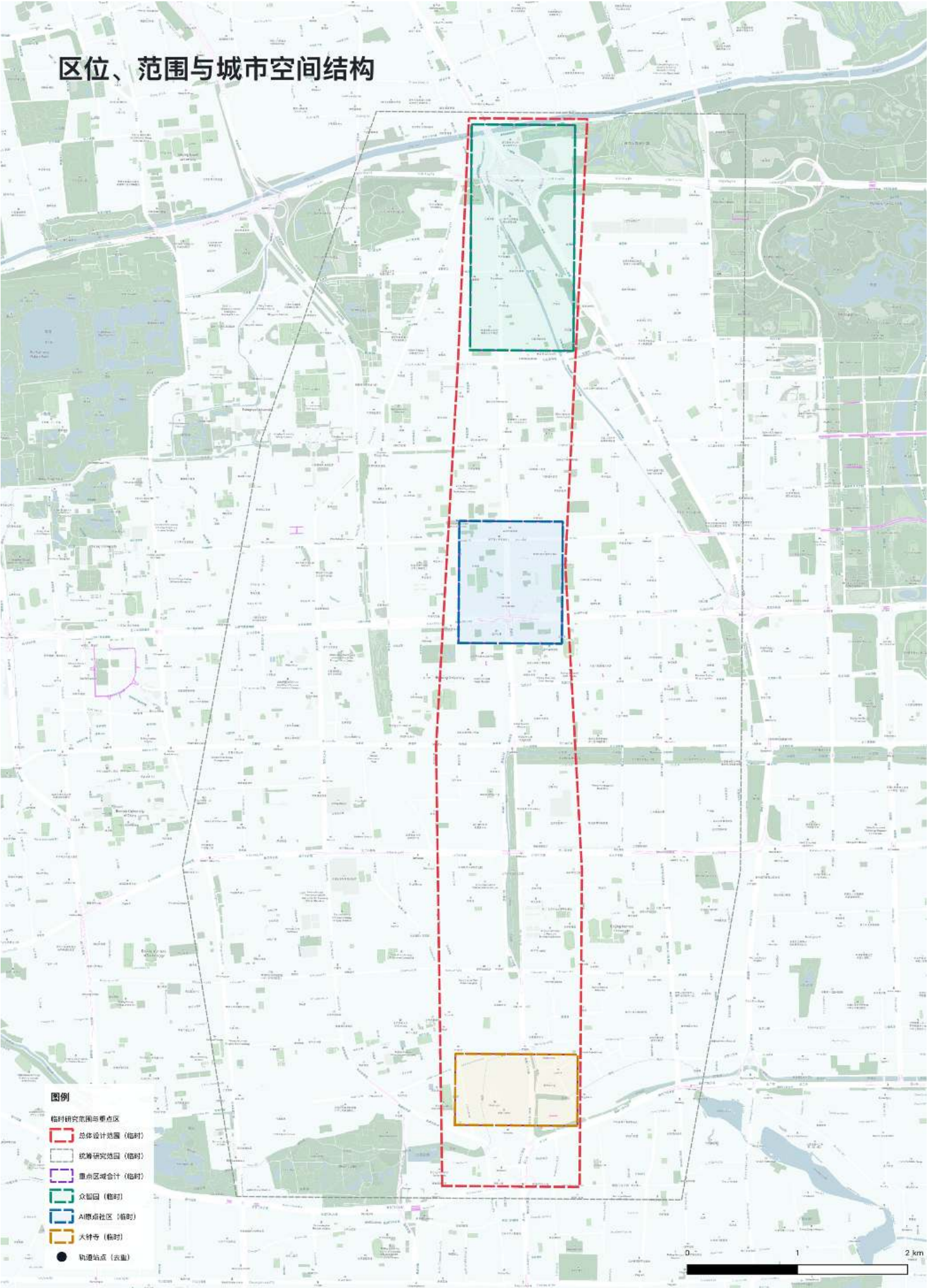
The three scales answer a continuous sequence of questions: how innovation resources become public capacity at the regional scale; how the heritage park and adjoining fabric form a complete urban structure at the overall-design scale; and how people enter, use, maintain and revise the ground in each key area. ^{Depth}

Scale	Question	Design work	Main output
Coordinated Research Area	How can universities, companies, communities and future AI form an open innovation ecosystem?	Study industry, knowledge, talent, service, accessibility and north-south differences	Analysis maps, core contradiction, case mechanisms and three-section strategy
Overall Design Area	How can the heritage park and adjoining ground become a continuous public city?	Continuous base, five accesses, Fifteen-Minute Open Units, public ground floors and blue-green mobility	Overall plan, integrated systems, renewal units and phasing
Key-Area Detailed Design	How do three innovation clusters create usable and operable public interfaces?	Refine functions, plans, sections, scenes and delivery conditions	Three plans, three sections, three scenes and project packages



Three scales and land-use structure

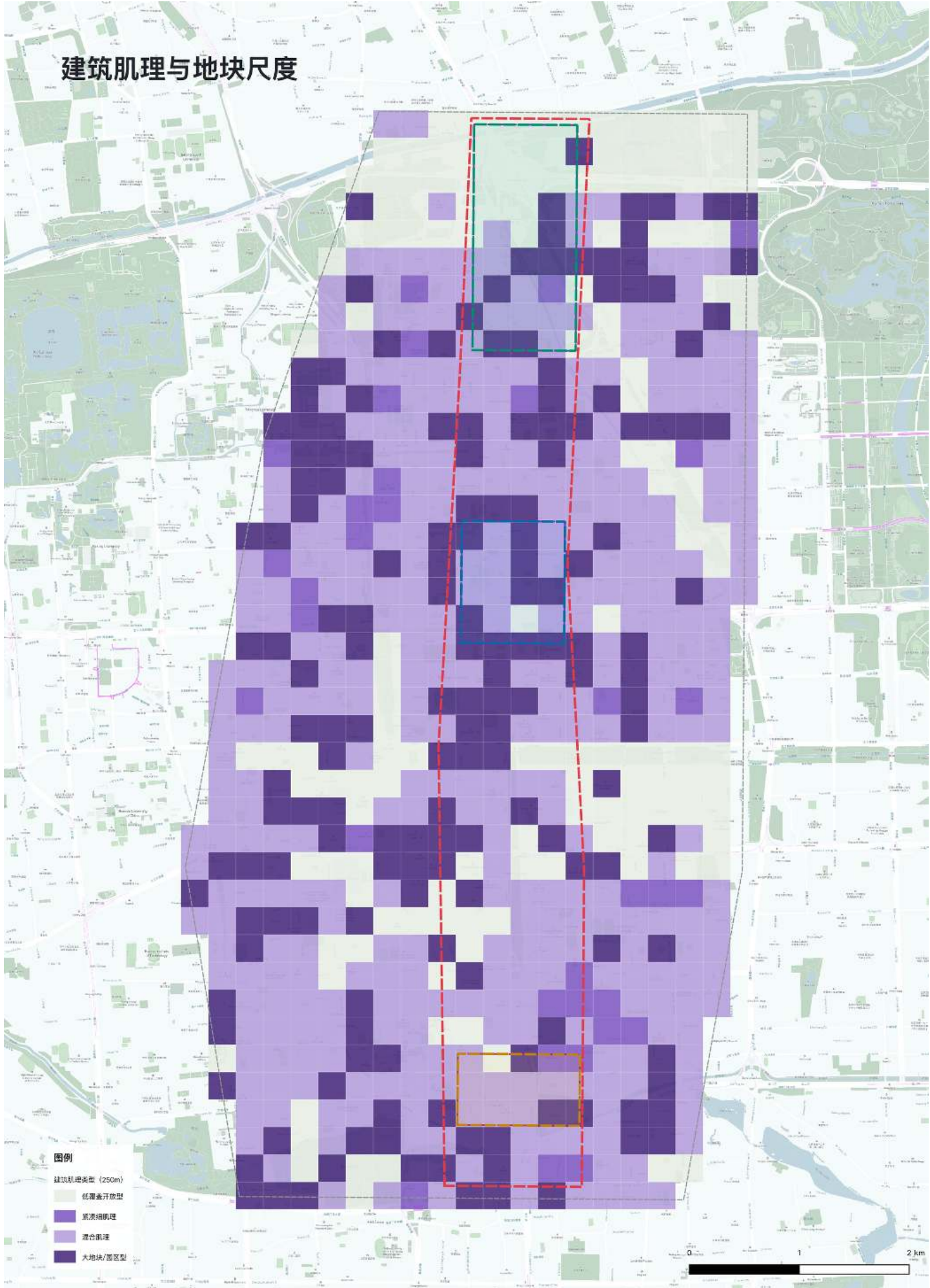
The three scales share one problem—space—delivery chain. Provisional boundaries are disclosed in the evidence package and every formal figure.



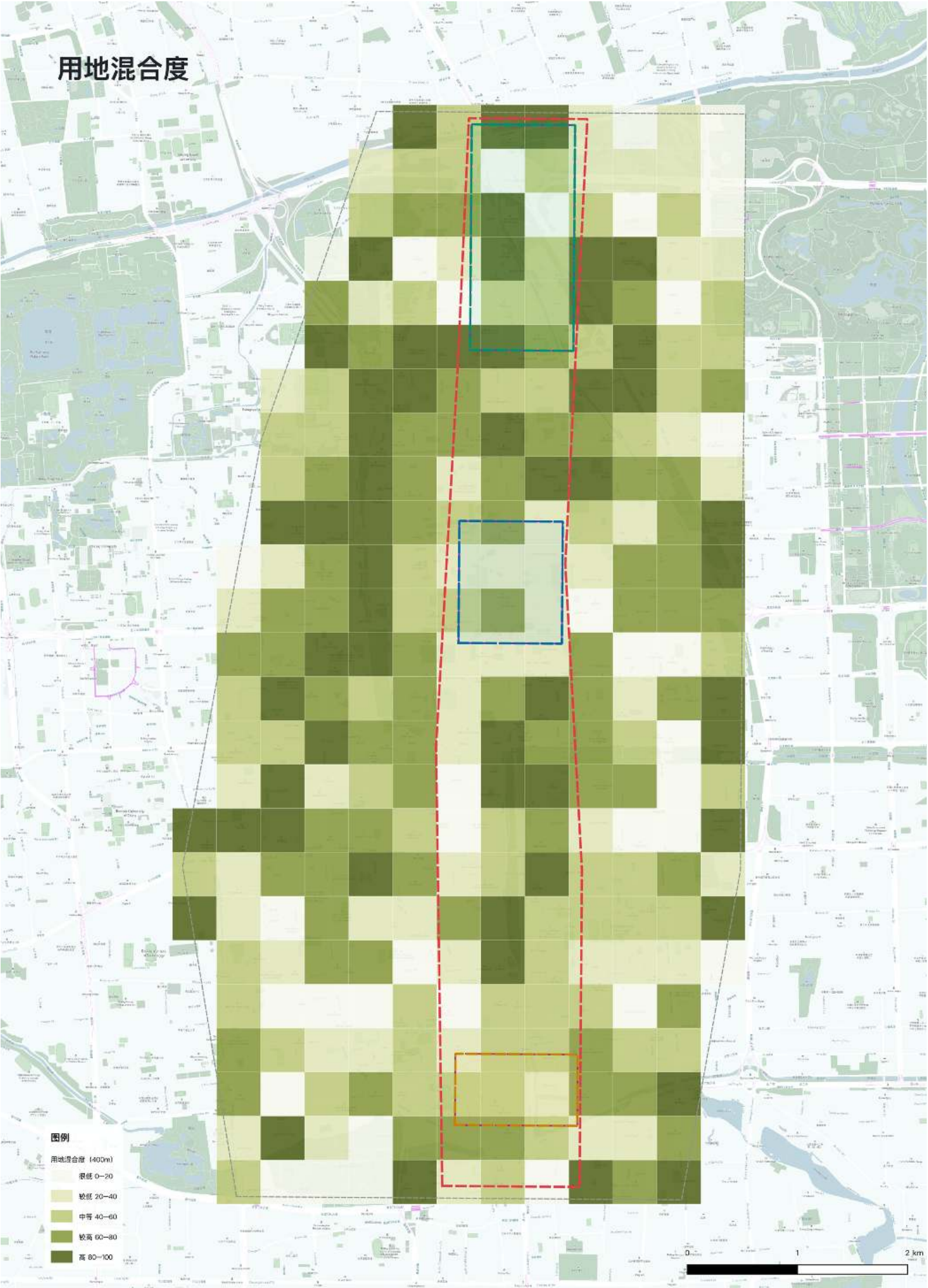
3 Coordinated Research Area: Industry and the Future City

3.1 Reading a Knowledge City through a Railway Corridor

The corridor extends from Xiaoyue River, Xuezh yuan and research parks in the north, through Liudaokou, Xueyuan Bridge and university/research clusters, to Zhichun Road, Dazhongsi, Jimen Bridge and Beijing North Station. Three urban fabrics coexist: northern parks beside blue-green resources; central university and institute superblocks interwoven with compact neighbourhoods; and southern communities, commerce, transit and railway gateways. A single renewal template cannot fit all three.



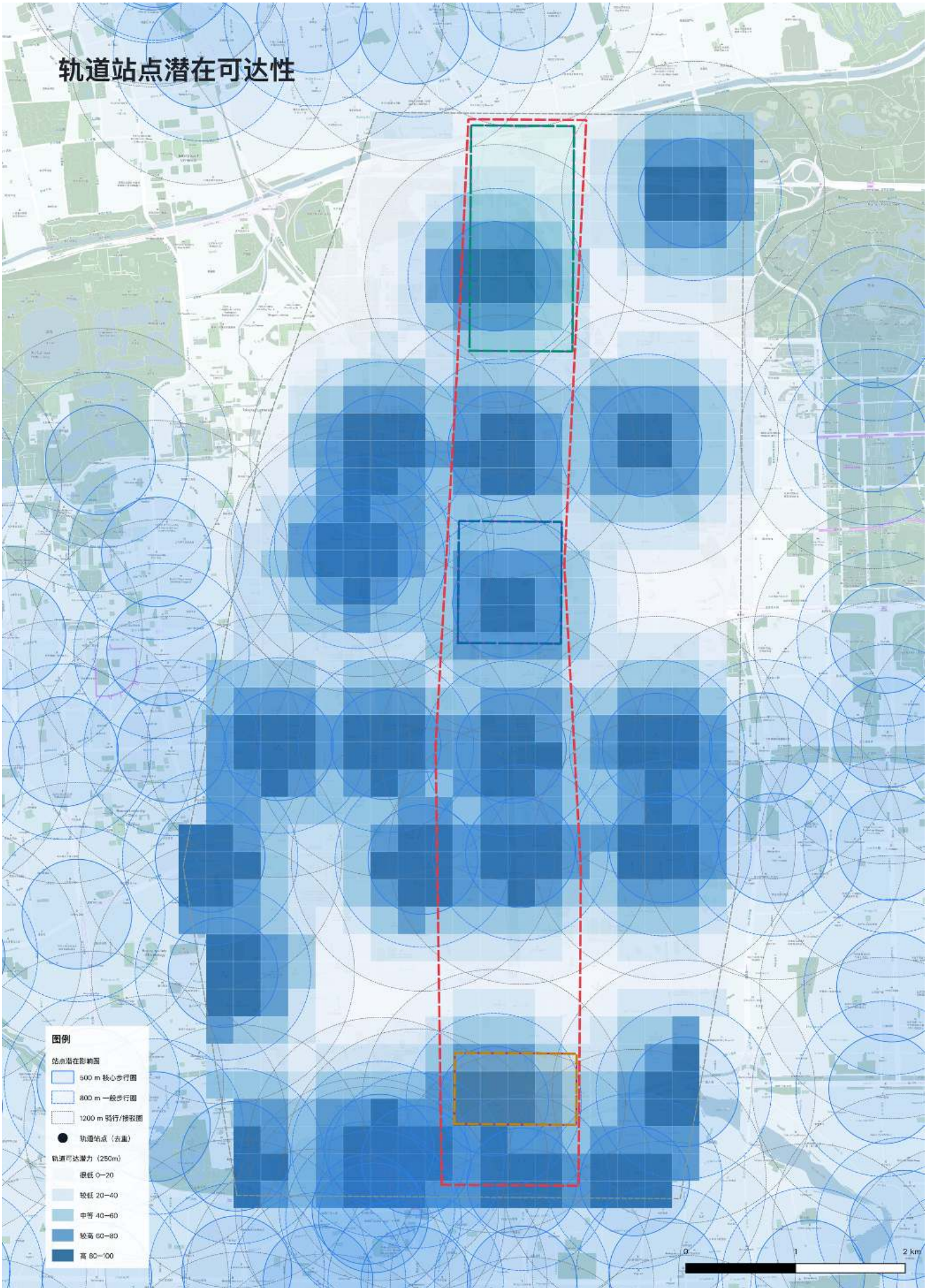
Coordinated Research Area: Industry and the Future City

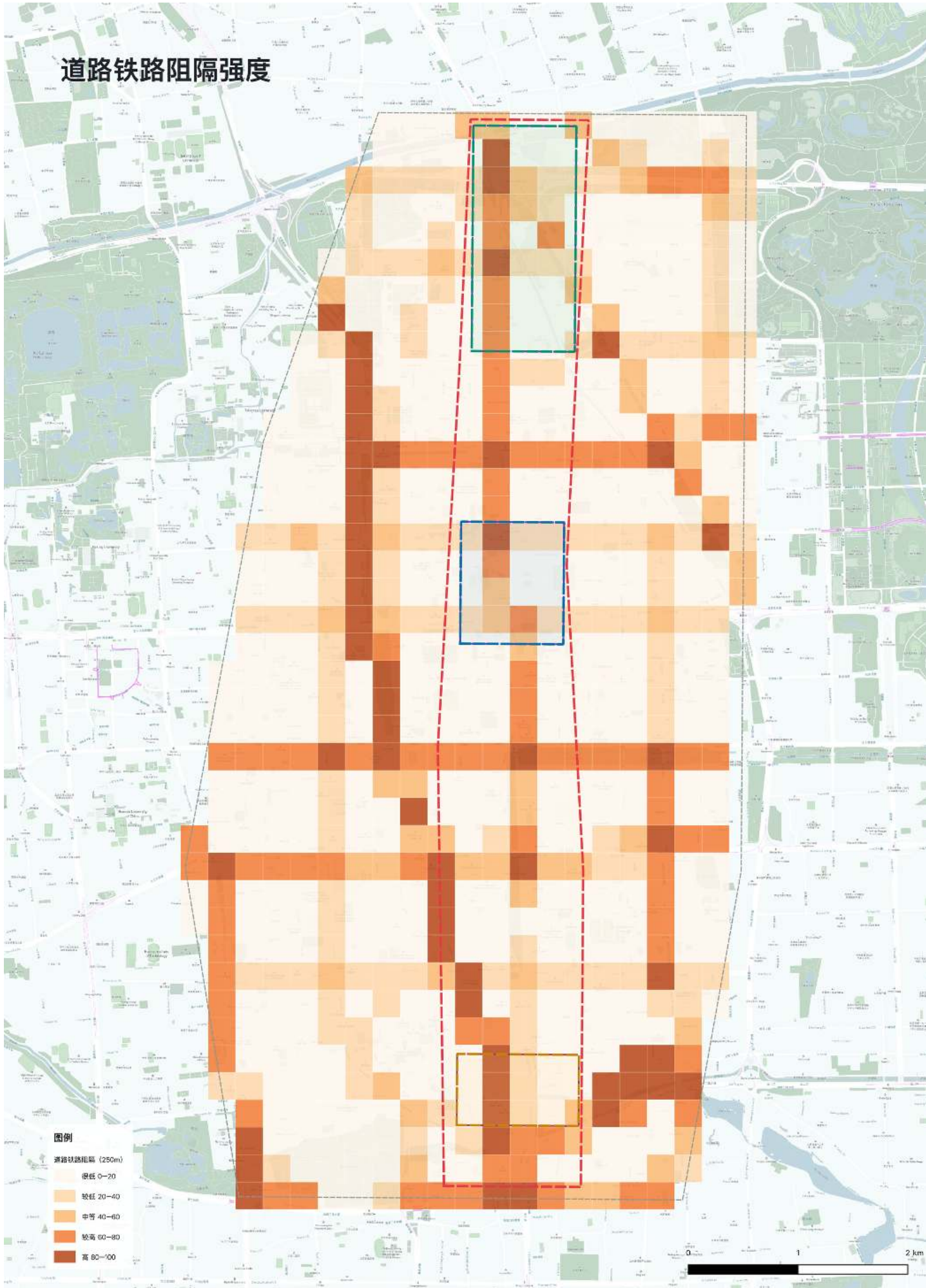
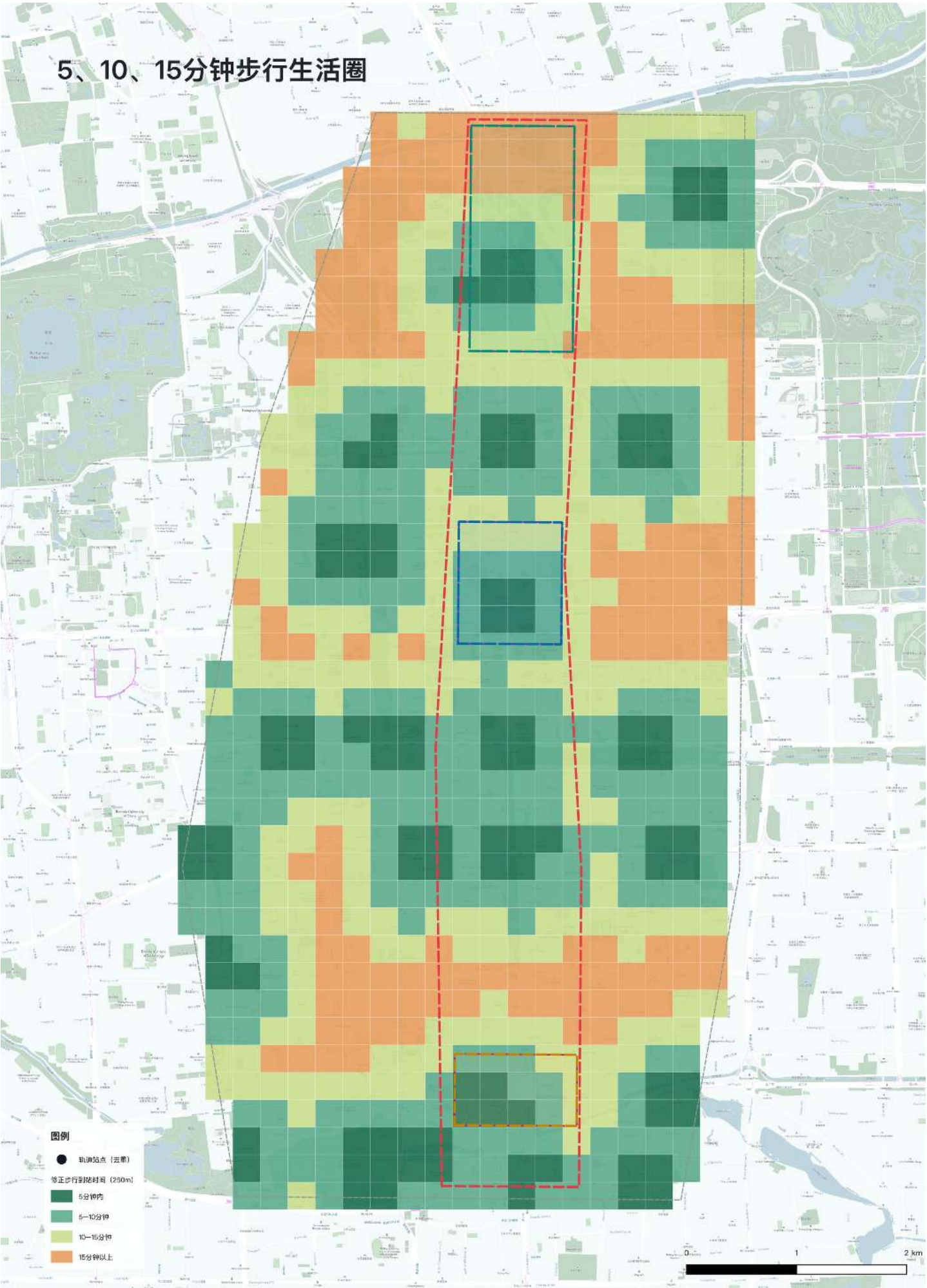


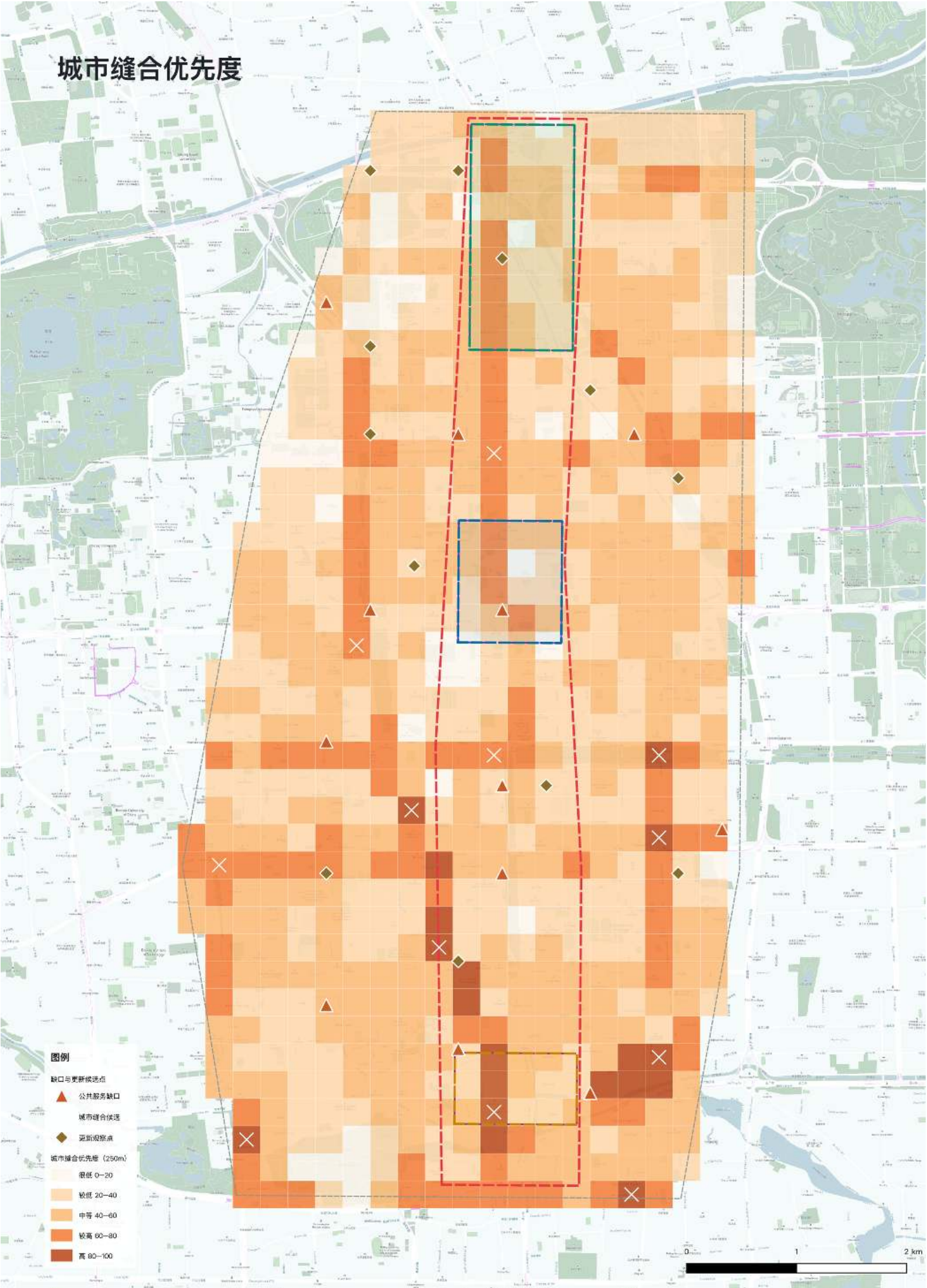
The corridor's potential comes from these systems coexisting; its difficulty comes from their separate rules and entrances. Knowledge production, technical testing, community service, rapid transit movement and park life sit close together geographically but remain out of phase operationally.

3.2 Good Transit Coverage, with the Last Link Deciding Real Access

Estimated five-, ten- and fifteen-minute walking coverage is 13.4%, 48.2% and 78.6%. Five-minute cycling reaches about 85.5% of the analysis grid; ten- and fifteen-minute cycling cover the full grid. Transit and cycling can support cross-district collaboration, while the last link from a station to a community, institutional entrance or public ground floor determines whether people can use the resource.

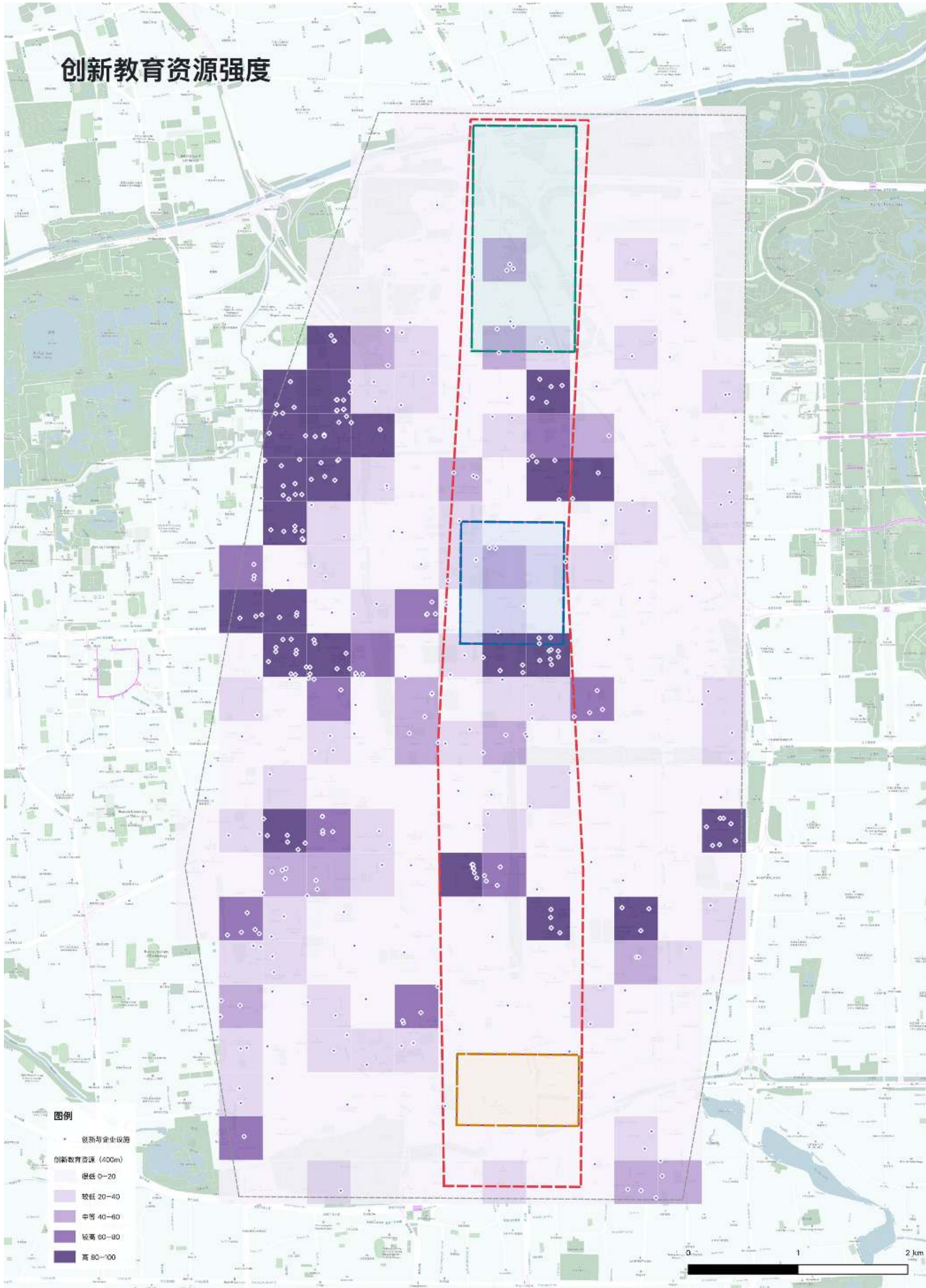
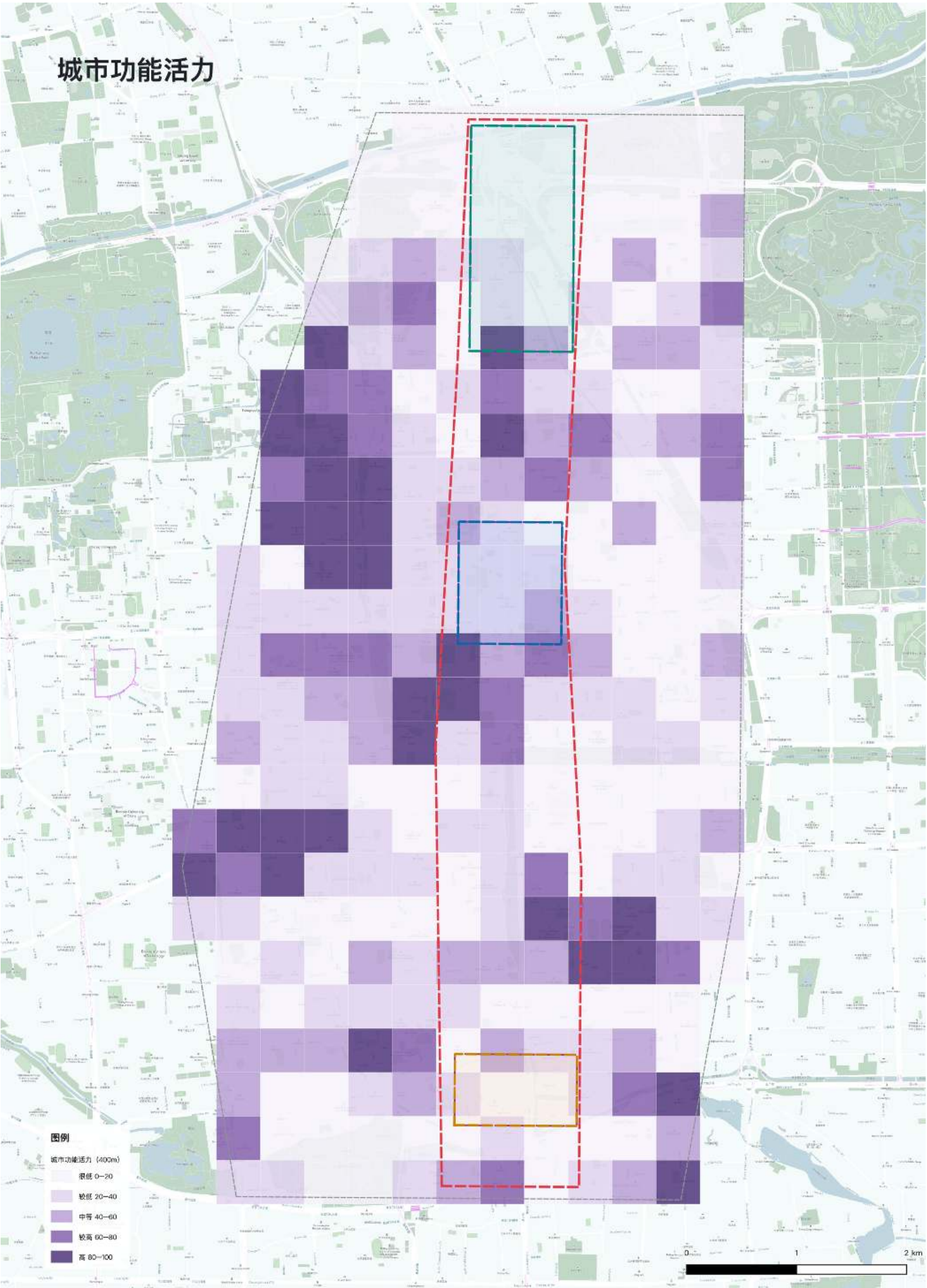


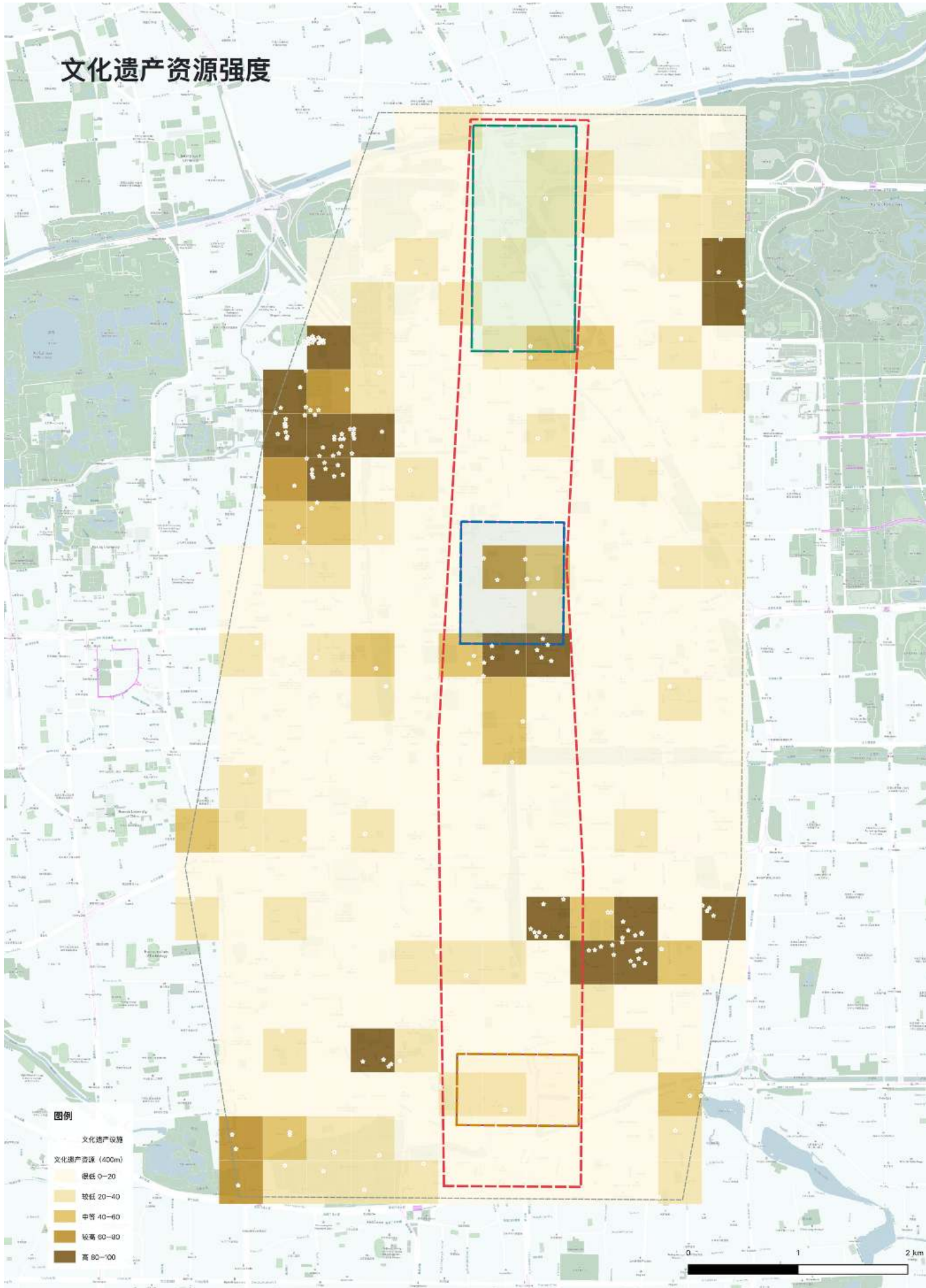
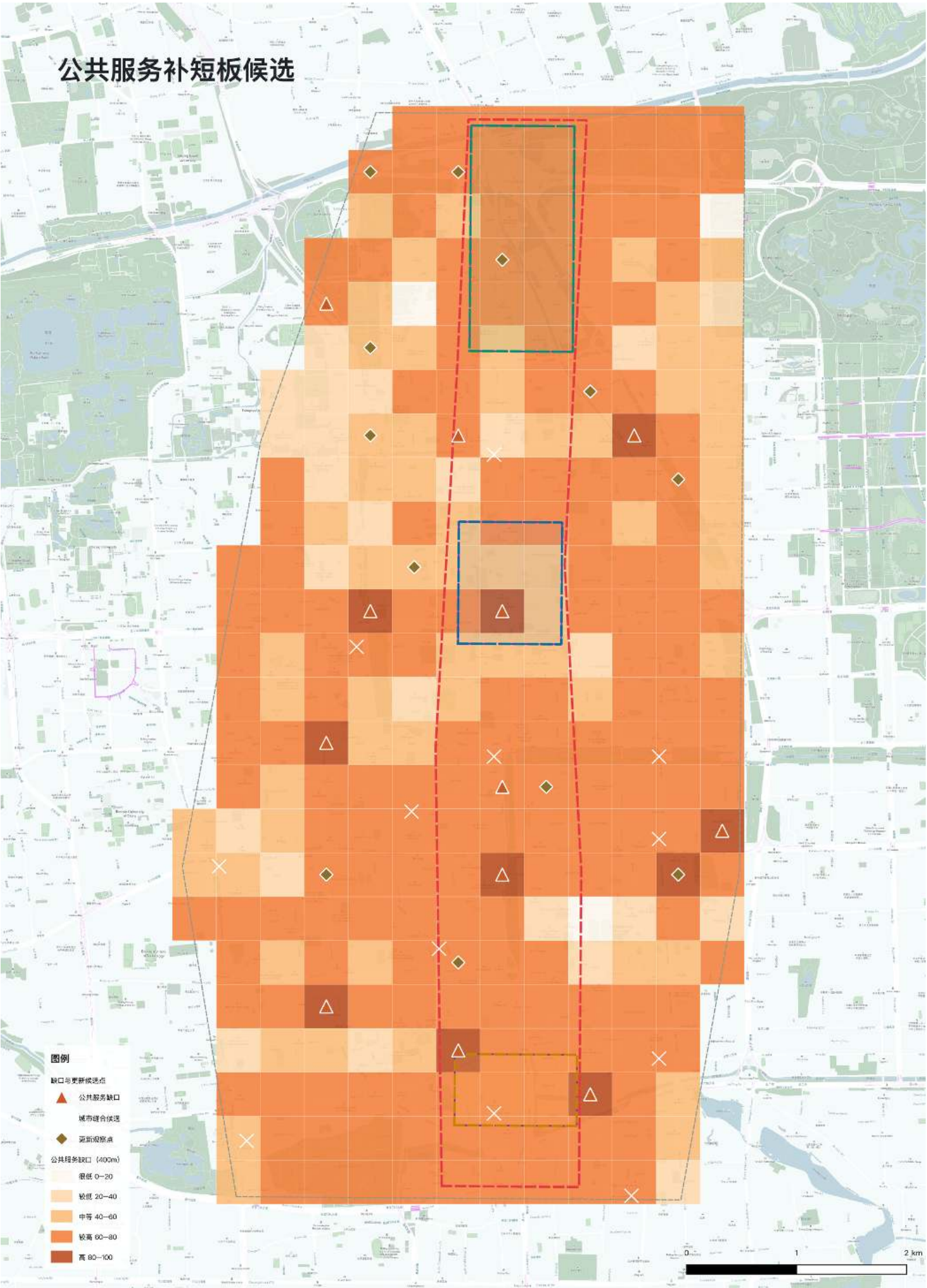




3.3 Concentrated Knowledge Resources, Thin Public Interfaces

Functional vitality, diversity and innovation/education resources form continuous high values around Xueyuan Road and central campuses; northern research parks and southern transit/commerce nodes add other clusters. Public interpretation, open courses, shared facilities, community-problem entry points and everyday service remain scattered.

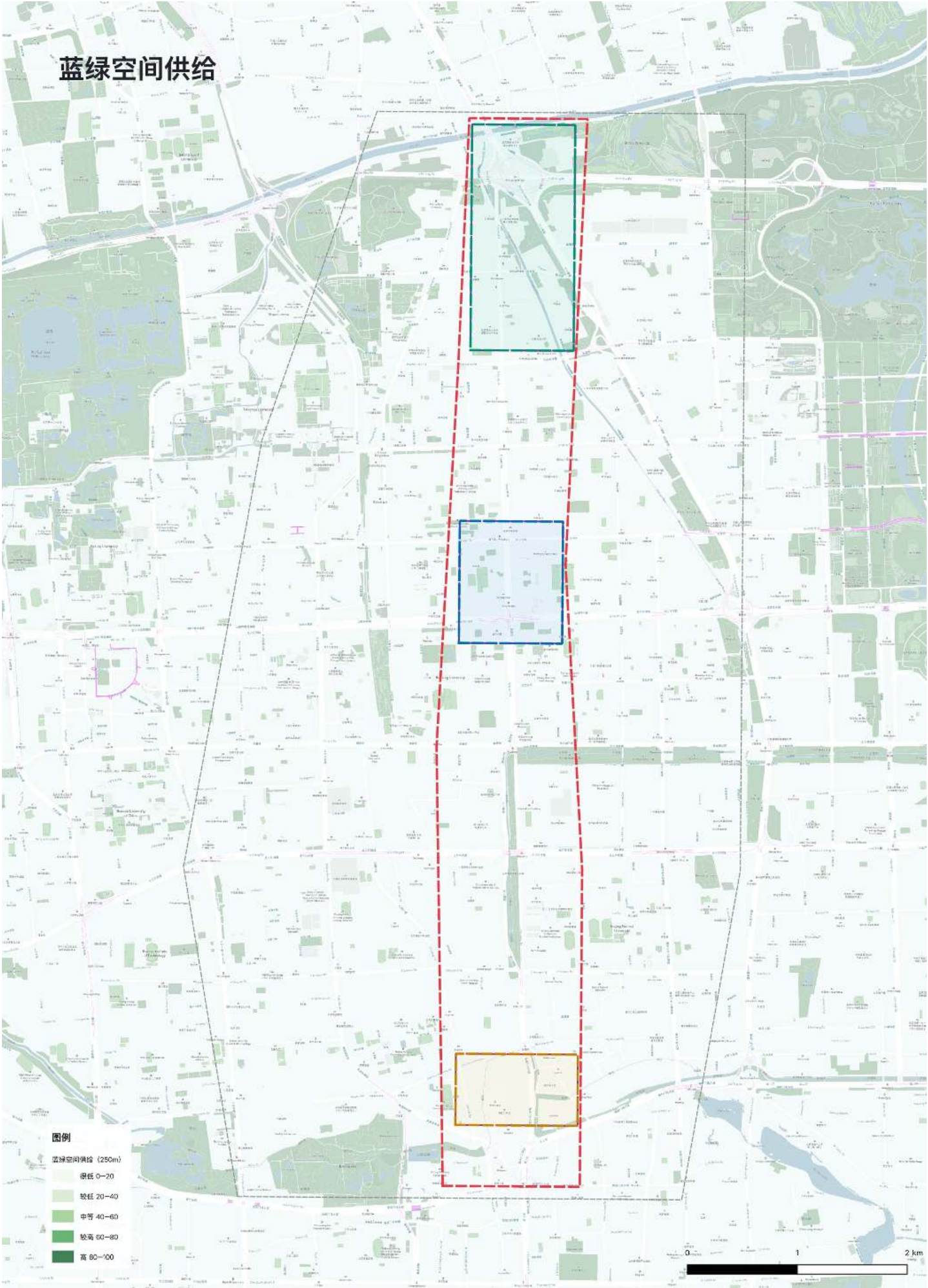


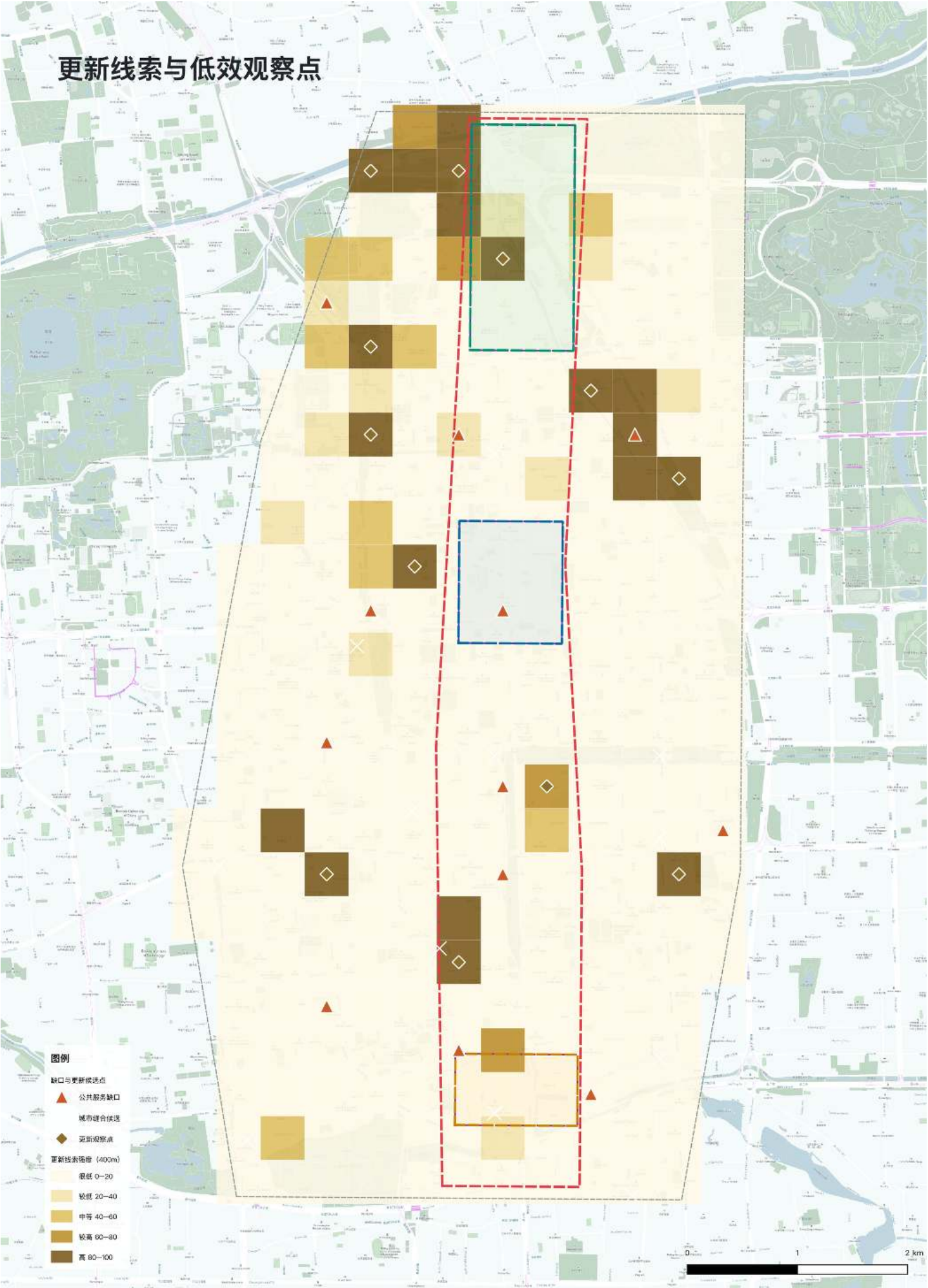


3.4 Blue–Green Resources, Renewal Clues and Three Distinct Key Areas

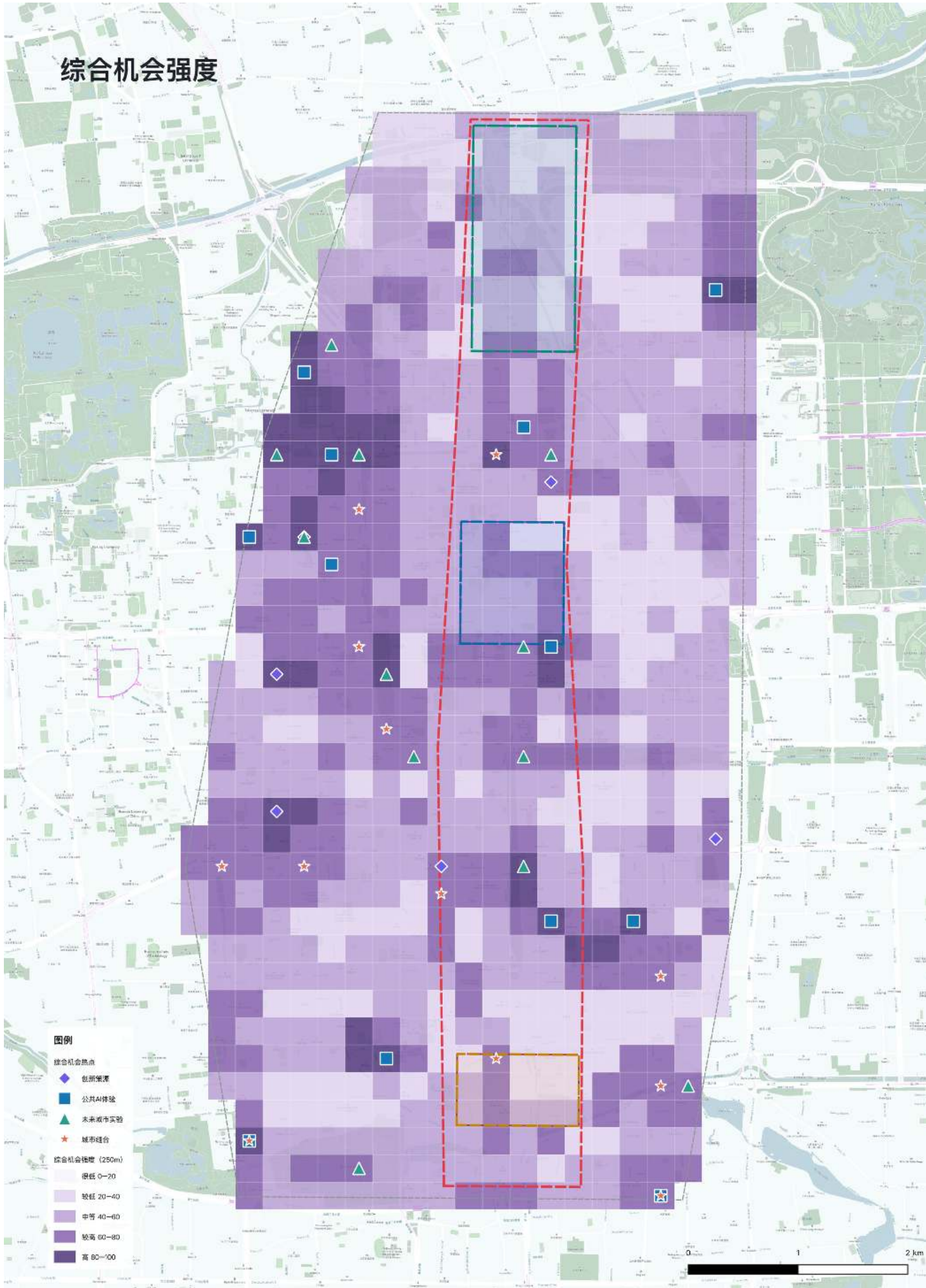
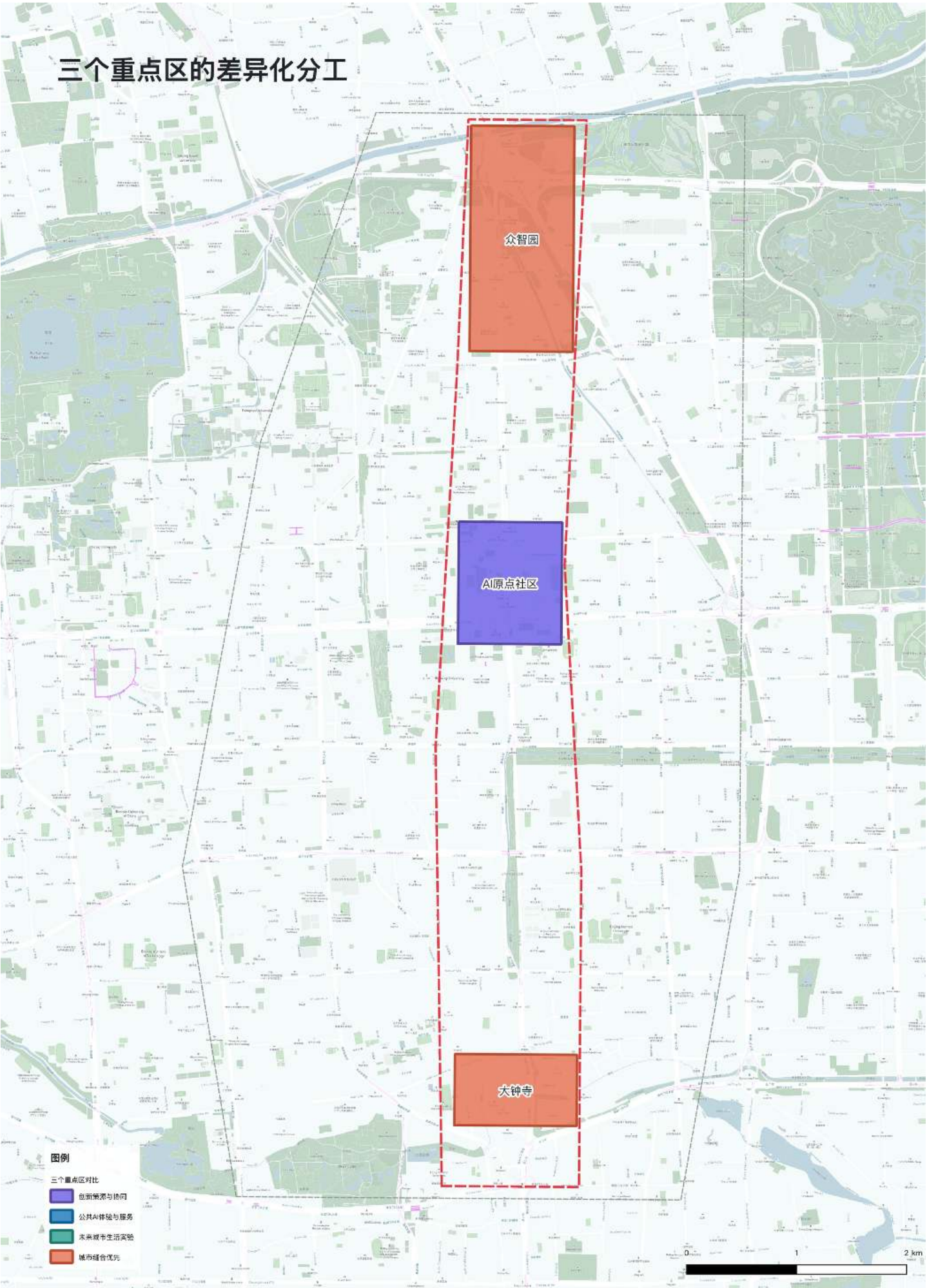
Xiaoyue River, northern parks and railway green space form a strong environmental base. Central green areas are divided by roads and institutional boundaries; the south requires trees, pocket spaces and rail–edge connections. Blue–green space must deliver stormwater, habitat, thermal comfort, recreation and environmental learning together.

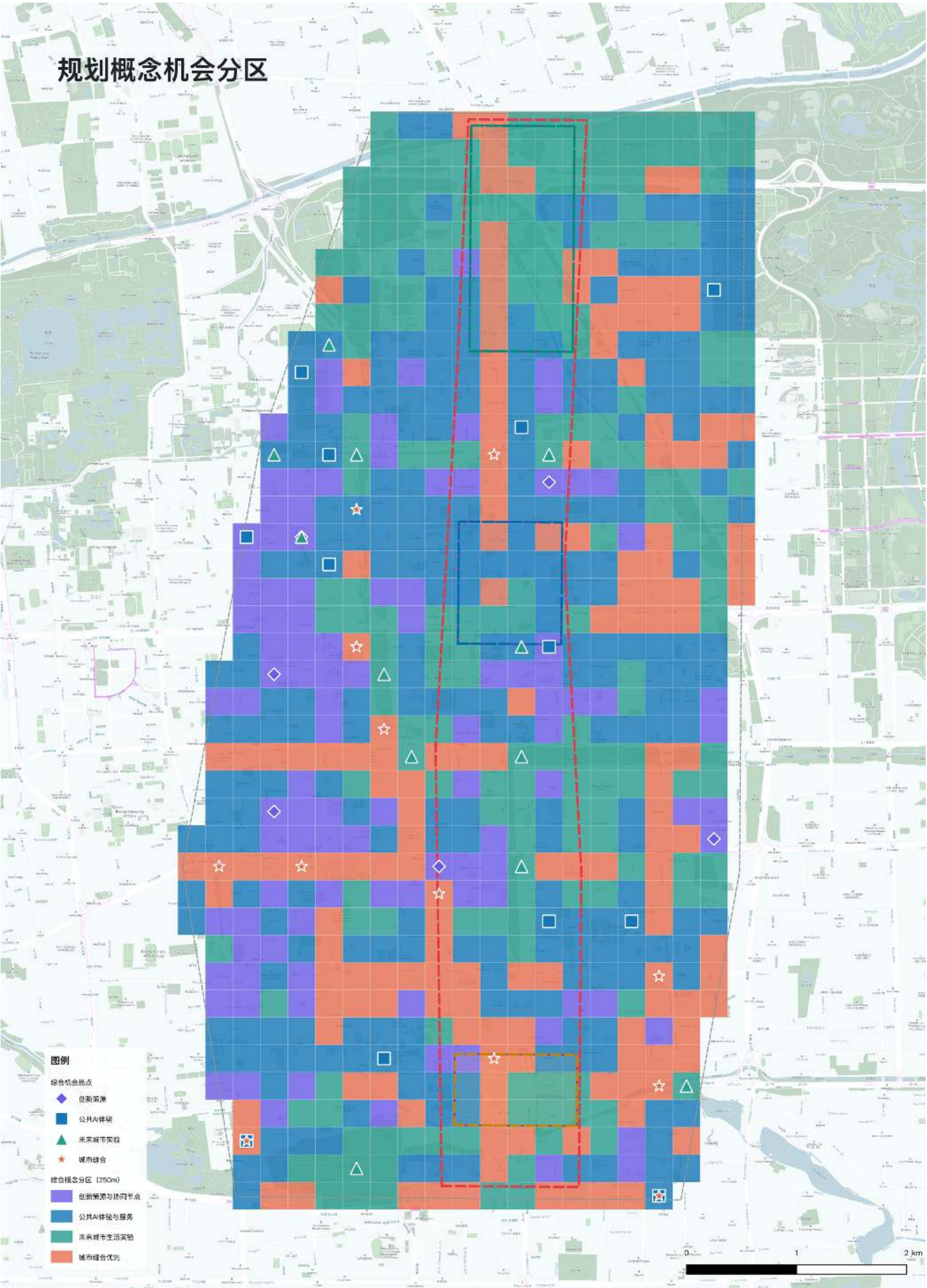
Much of the first public transformation can begin in existing space: open institutional edges, convert ground floors, improve transverse streets, add forecourts and reorganise logistics. Demolition decisions await parcel–by–parcel evidence on ownership, structure, fire safety, heritage and current use.





Zhongzhiyuan can connect station, park and river as environmental-learning ground. AI Origin can compile research into public knowledge through dense public ground floors. Dazhongsi can orchestrate transit, employment, health and culture for a high-flow established district.





3.5 The Core Contradiction

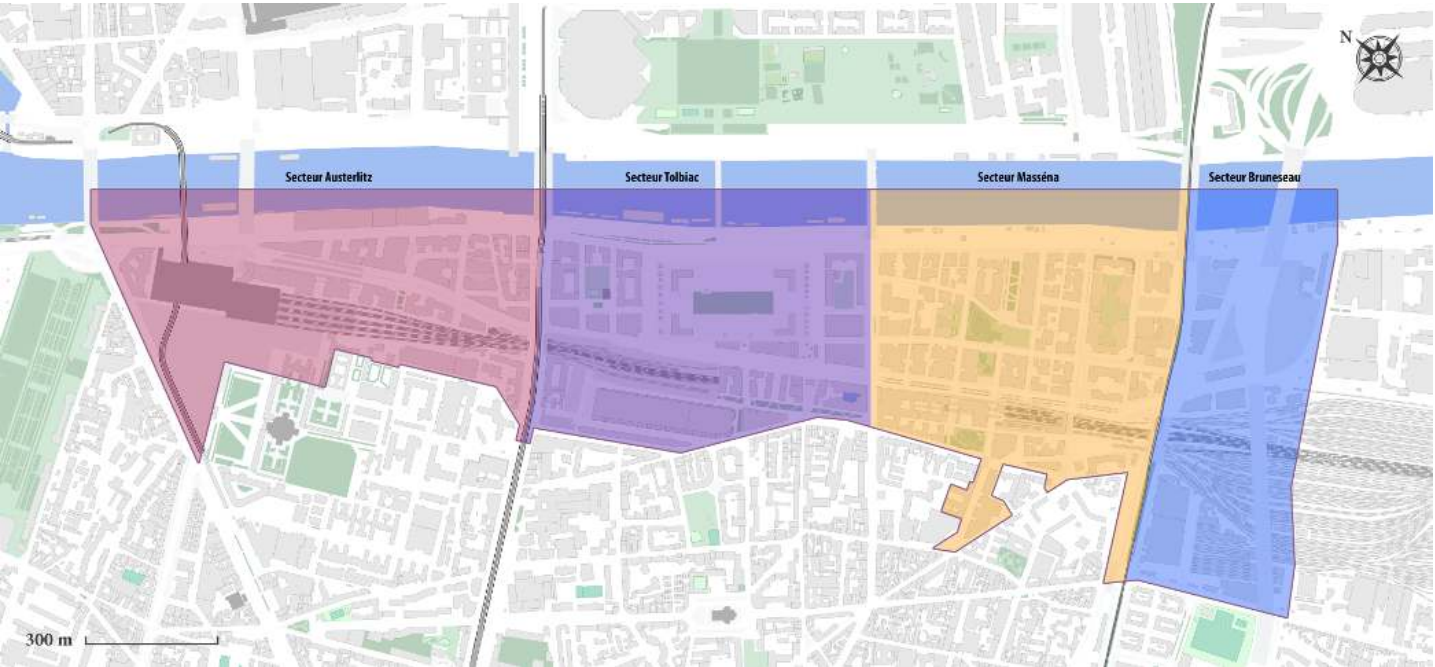
Five findings reinforce one another: innovation and education resources are highly concentrated; activity and public service form local high values; transit and cycling provide broad potential coverage; transverse barriers and superblocks create real detours; and public ground floors and continuous blue-green interfaces remain thin. The core contradiction is therefore: **innovation resources are intensely concentrated, while interfaces for public transformation remain broken.**

The design focuses on interfaces. Knowledge enters public ground floors through courses, models and explanation; technology enters public space through bounded testing and human service; urban problems enter research systems through community points and open lists; parks, rivers and streets enter daily life through continuous paths. Each opening has a time, permission, accountable operator and maintenance commitment.

3.6 How Cases Convert Concentrated Resources into Public Capacity

Paris Rive Gauche: Bridging a Live Railway with City Streets

Paris Rive Gauche divides a live railway bundle into implementable sectors and restores continuity through bridge streets, facilities, open space and long-term public development. The Jing-Zhang translation is five transverse project packages, with ground improvements, entrance opening and future structural crossings advanced on separate timelines. [Source](#) [Source](#)



Paris Rive Gauche implementation sectors
Four implementation sectors, 2022. Cartography: Paris 16; basemap: OpenStreetMap contributors; CC BY-SA 4.0; rasterised from Wikimedia without alteration.

Coordinated Research Area: Industry and the Future City

Yangpu Riverfront: A Continuous Public Space Linking Heritage, Community and Everyday Operations

The southern Yangpu Riverfront combines continuous walking, transverse passages, heritage nodes, phased renewal and sustained operations. Jing-Zhang transfers this as a continuous base plus frequent entrances, heritage learning and everyday programming. [Source](#) [Source](#)



Yangpu Riverfront masterplan

The 5.5 km southern Yangpu Riverfront masterplan. Yihao Wu and Yang Liu, Figure 5, Sustainability 15(6):5060, 2023; CC BY 4.0.



Yangpu Riverfront at night

Night-time walking space, 2019. Ding Kezhong; CC BY-SA 4.0; Wikimedia Commons.

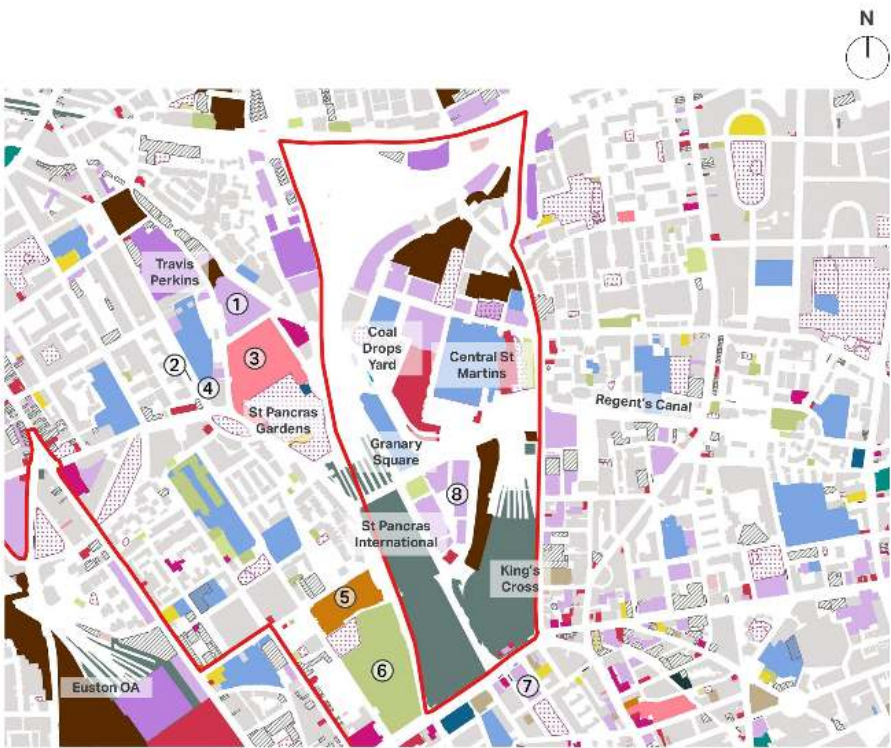
King's Cross: A University Anchor, Public Space and Heritage Reuse

King's Cross combines a university anchor, research, commerce, housing, heritage reuse and high-quality public space through long-term development and operation. Jing-Zhang translates this into institutional public entrances, shared forecourts, knowledge service and independent evening operation. [Source](#) [Source](#)

King's Cross Land uses

Knowledge Quarter

- Strong education focus that acted as catalyst for placemaking, including Central St Martins.
- Life Sciences cluster just outside of the red line boundary (Crick Institute, emerging life sciences scheme in York Way and in Tribeca).
- Attraction and Activity (Granary Square, Gasholder Park, Pancras Square).



Source: OSNGB, City Intelligence Unit (GLA)

MedCity, Knowledge Quarter:

- | | | | |
|----------------------------|---|-------------------------------|------------------------|
| ① Tribeca | ③ Camden and Islington NHS foundation trust | ⑤ The Francis Crick Institute | ⑦ MSD Discovery Centre |
| ② Royal Veterinary College | ④ London BioScience Innovation Centre | ⑥ Alan Turing Institute | ⑧ Astra Zeneca HQ |

King's Cross land-use structure

Land-use structure from the Greater London Authority Opportunity Area Portrait, OGL v3.0; cropped to the map frame.



Granary Square, King's Cross

Granary Square, 2019. Christian Michel; CC BY 2.0; Wikimedia Commons.

22@, KIC and Kendall Square: Bringing Knowledge Institutions into Streets and Communities

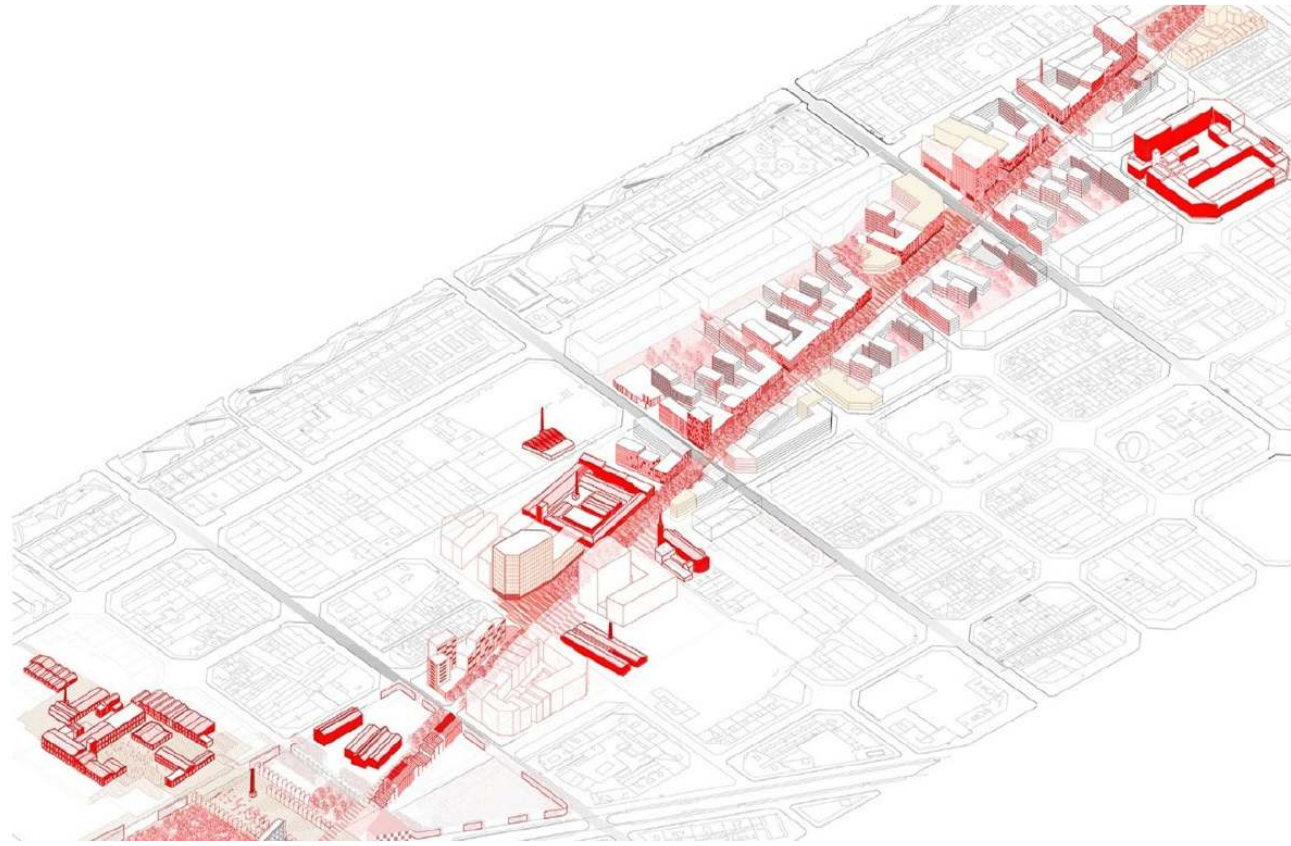
KIC, 22@ Barcelona and Kendall Square add knowledge–street, infrastructure–investment and innovation–space mechanisms. Their key lesson is to combine specialised research capacity with public streets, public ground floors, daily commerce and transparent long–term governance.



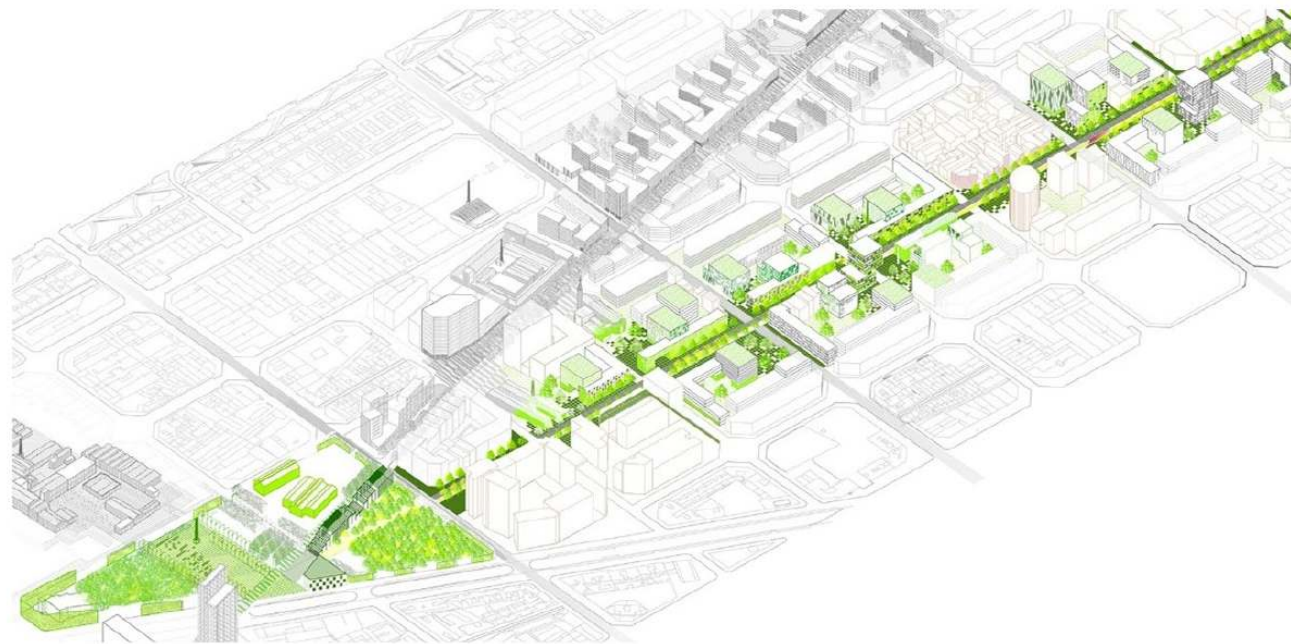
Pere IV in 22@ Barcelona

Pere IV, 2016. Canaan; CC BY-SA 4.0; Wikimedia Commons.

Coordinated Research Area: Industry and the Future City



Pere IV green-axis proposal
Green-axis proposal, Barcelona Regional, 2020; CC BY 4.0; cropped from the official planning document.



Cristóbal de Moura green-axis proposal
Green-axis proposal, Barcelona Regional, 2020; CC BY 4.0; cropped from the official planning document.



Community ice rink in Kendall Square
Community ice rink, 2014. Caroline Culler; CC BY-SA 4.0; Wikimedia Commons.

Coordinated Research Area: Industry and the Future City

Extended Cases: Linear Parks, Urban Rooms, Blue–Green Trials and Long–Term Operations

The wider set adds Gyeongui Line Forest Park, The Bentway, Stockholm Royal Seaport, Shougang Park, Guiwan Park and Hong Kong Science Park.

Case	Transferable mechanism	Jing–Zhang placement
Gyeongui Line Forest Park	A multi–entrance linear park with everyday community use	Dense entrances from the heritage park to transverse streets
The Bentway	Reversible urban rooms under infrastructure, seasonal events and operational organisation	Time–shared public space under bridges, junctions and infrastructure edges
Stockholm Royal Seaport	Blue–green indicators, site conditions, compliance ledgers and urban trials	Zhongzhiyuan environmental learning, corridor–wide versioning and post–occupancy review
Shougang Park	Industrial–heritage reuse, technology industry and event operations	Railway heritage nodes, adaptive building reuse and cultural programming
Qianhai Guiwan Park	Stormwater, habitat, public activity and digital maintenance combined	Xiaoyue River, the railway park and street rainwater space
Hong Kong Science Park	Phased research premises, collaboration space, talent services and waterfront opening	R&D–public interfaces and operational services in Zhongzhiyuan and AI Origin

Together the twelve cases yield four stable conclusions: a continuous longitudinal base needs dense transverse entrances; knowledge institutions need public ground floors and everyday uses; different sections can share open standards; and long projects need phased responsibility, maintenance funding and operational ledgers.

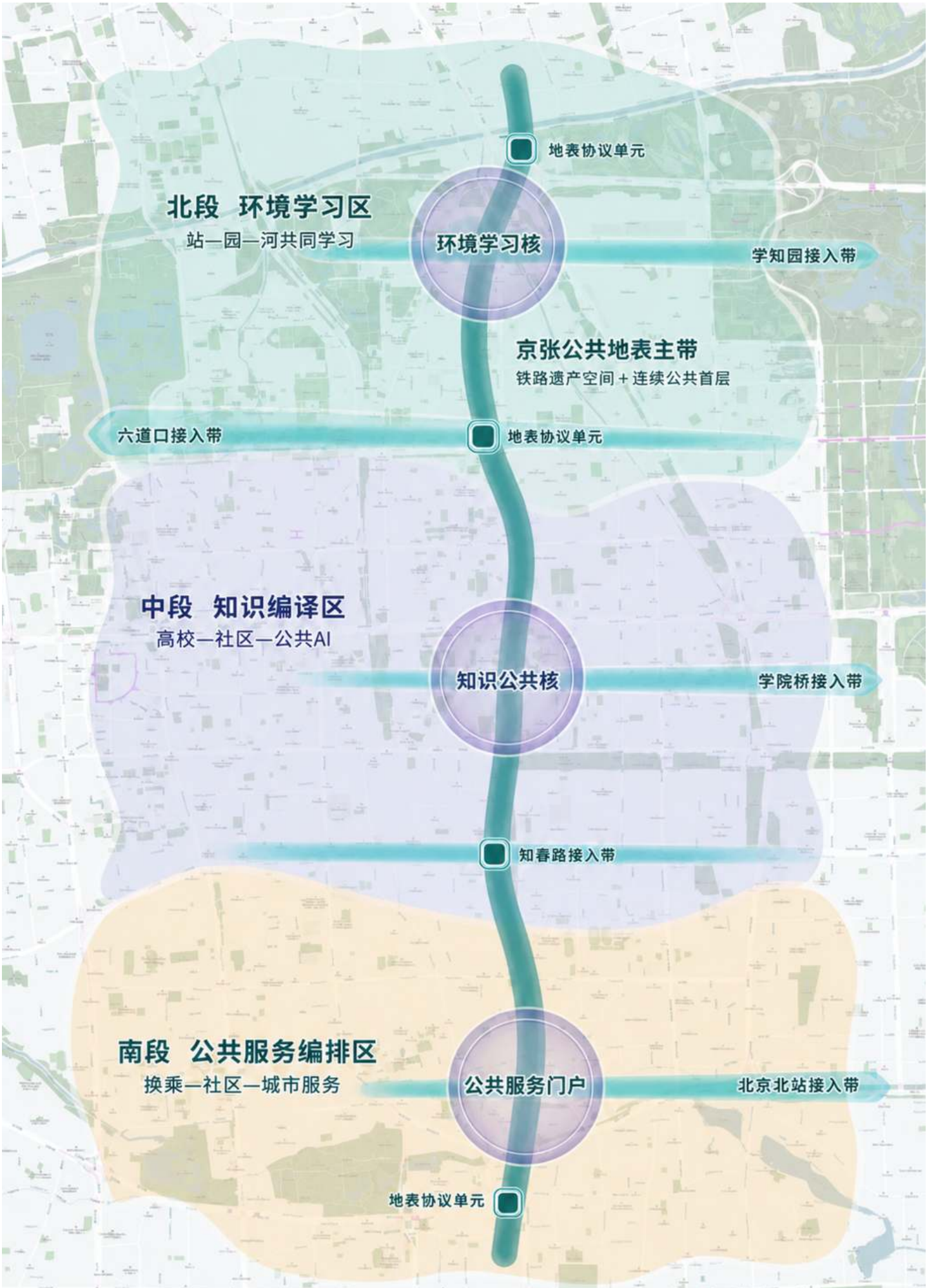
4 Overall Design Area: Urban Renewal and Urban Design

4.1 The Spatial Prototype of Open-Source Ground

Open-Source Ground extends open-source principles from code to urban space, knowledge, data and models, facilities, urban problems, public services and governance interfaces. Each public interface answers six questions: what is open, how it is reached, when it is usable, how one contributes, who revises it, and how one exits. Entrance, time, permission, responsibility, version and recovery form a Ground Protocol Unit. ^{Depth}

Public right	Spatial action	Everyday result
See	Make entrances, open content, status and responsibility legible	People understand conditions before arrival
Reach	Continuous accessibility, primary/backup routes and station links	Communities, transit, parks and institutions form complete paths
Use	Tier basic access, time-sharing, collaboration and controlled contribution	Everyday and specialised access have clear thresholds
Contribute	Urban-problem entry points, courses, models, tools and local experience	Real problems enter research and public service
Modify	Public evaluation, appeal, correction and version records	Rules adjust using evidence from use
Exit	Data deletion, equipment shutdown, activity closure and spatial recovery	Public routes and essential services survive technical change

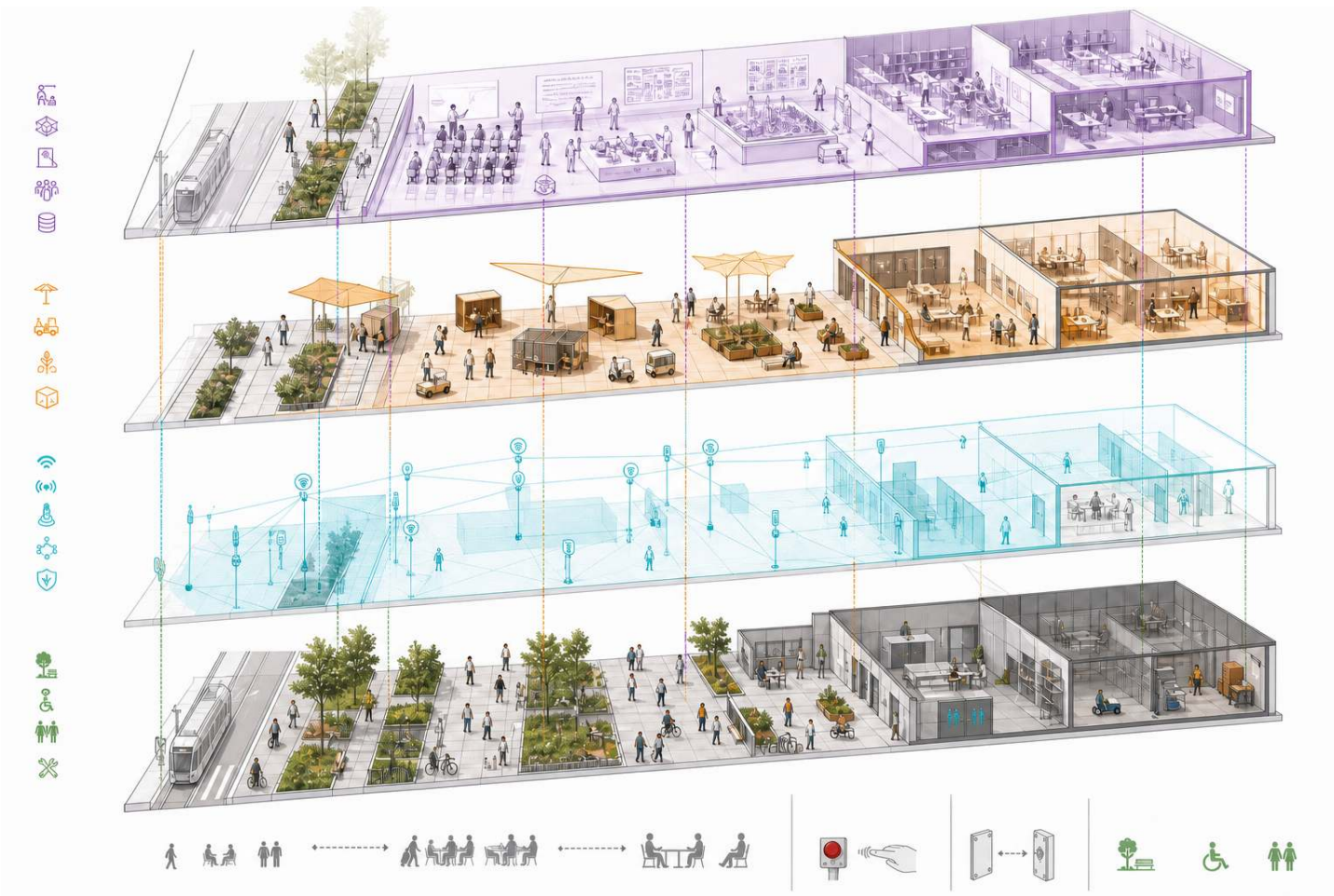
Five operations make these rights spatial: open institutional edges; thicken the public ground through park, forecourt and ground floor; establish transverse access; embed human service, tools, models and infrastructure; and manage temporary layouts, hours and technical modules by version.



Overall Design Area: Urban Renewal and Urban Design

4.2 Four Grounds in the Same Space

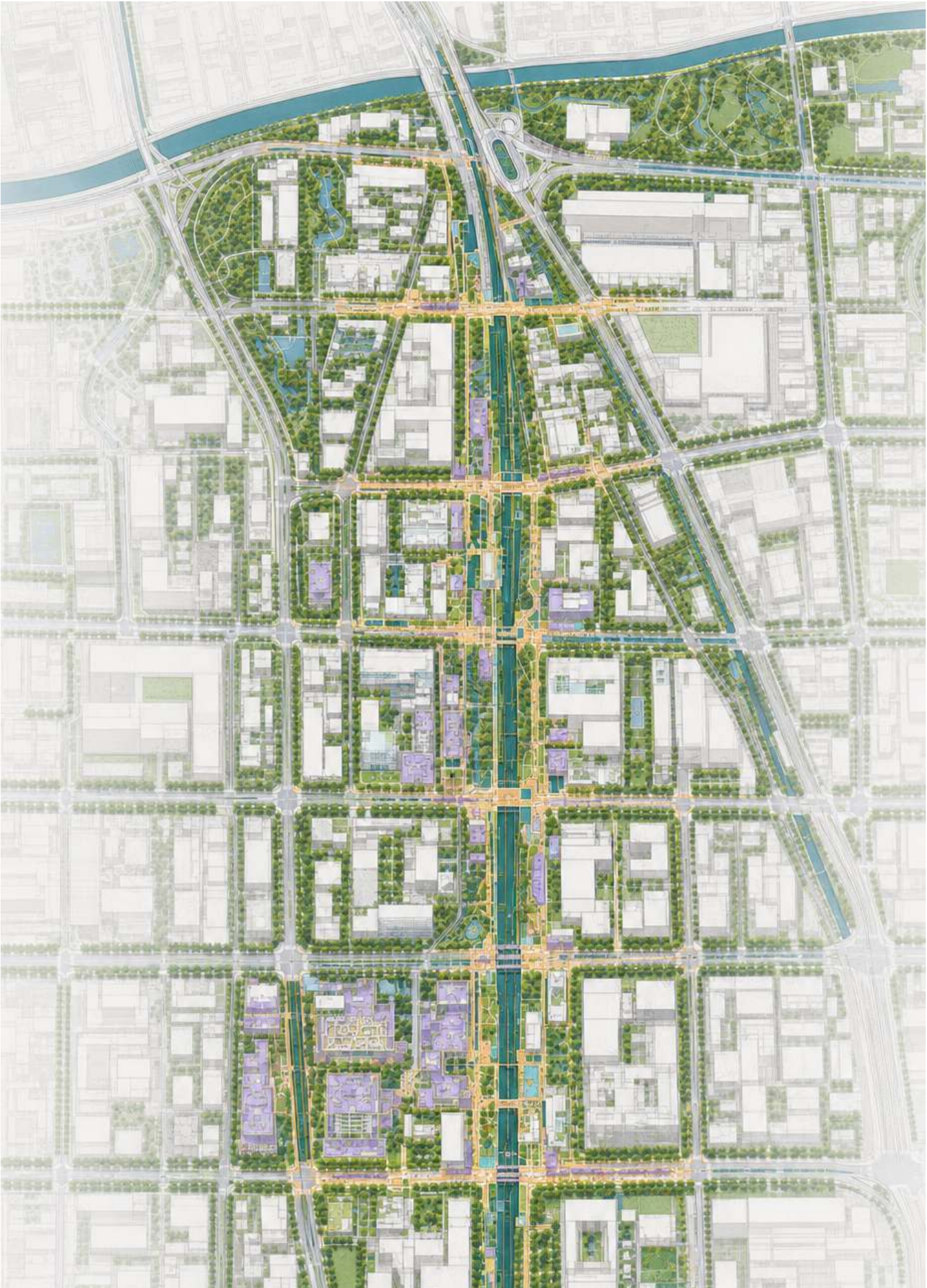
Physical ground provides walking, cycling, public ground floors, landscape, shade, sanitation, storage and logistics. Digital ground discloses status, environment, capacity, energy, sources and error. Generative ground assists bounded adjustments to shade, irrigation, lighting, activity layouts and service. Knowledge ground turns courses, research, community experience, maintenance and failure into public resources.



Four-layer ground section
Controlled research space, public ground floor, shared forecourt, walking/cycling and blue-green space form the physical base; digital, generative and knowledge layers occupy the same spatial unit.

4.3 One Continuous Public Ground and Five Transverse Accesses

The Jing-Zhang Railway Heritage Park forms the continuous base, supported by parallel streets, park paths and community routes. Five accesses connect Xuezhiyuan—Xiaoyue River, Qinghuadongluxikou—Liudaokou, Xueyuan Bridge—Al Origin, Zhichun Road—Dazhongsi—Jimen Bridge, and Beijing North Station—Gaoliang Bridge. Each is an implementable renewal package combining mobility, ground floors, landscape, public service and operation.



开源地表 | 总体空间结构核验

统一视图116.285-116.400E、39.915-40.055N | 完善南北走廊与必要外围



总体结构：北段环境学习 / 中段知识编译 / 南段公共服务 / 三个重点区 / 连续公共地表 / 五条横向接入 / 五个重点空间单元

Overall Design Area: Urban Renewal and Urban Design

4.4 Fifteen–Minute Open Units and Key Spatial Units

Five–minute provision covers movement, rest, water, sanitation, human information and help; ten–minute provision adds public ground floors, repair, learning and care; fifteen–minute provision connects courses, workshops, urban models, river learning and cross–institutional collaboration. Ground Protocol Units manage rules, Fifteen–Minute Open Units manage service coverage, and Key Spatial Units receive integrated design.

Key Spatial Unit	Main intervention	Overall role
Xuezhiyuan—Xiaoyue River	Station—park—river routes, public ground floors, ecological learning and maintenance	Northern environmental learning; Zhongzhiyuan
Qinghuadongluxikou—Liudaokou	Two–station connection, campus gateways, evening paths, cycling and equipment transfer	Central transverse stitching
Xueyuan Bridge—AI Origin	Public ground–floor cluster, urban model room, forecourts and community service	Central knowledge compilation; AI Origin
Zhichun Road—Dazhongsi—Jimen Bridge	Four potential access directions, continuous cycling and community/commercial ground floors	Southern service orchestration; Dazhongsi
Beijing North Station—Gaoliang Bridge	Accessible transfer, community passage, visitor service and railway interpretation	Southern railway gateway



Qinghuadongluxikou—Liudaokou transverse access
A transverse public route links two stations, university parks, communities, commercial ground floors and heritage space. Exact exits, crossings and clearances await specialist survey.

Overall Design Area: Urban Renewal and Urban Design

4.5 Three Ground Prototypes

Section	Resource and problem	Prototype	Main action
North	Research, parks and river are close; station—park—river continuity is weak	Environmental—learning ground	Link station, park, community and river; add environmental courses, tools and controlled equipment routes
Centre	Dense universities, AI companies and communities lack a common interface	Knowledge—compilation ground	Cross—institutional access, public ground—floor clusters, urban models and evening learning
South	Transit, communities, commerce and railway gateway overlap; services are fragmented	Public—service—orchestration ground	Potential four—direction access, continuous accessibility/cycling, service ground floors and gateway functions

5 Key–Area Detailed Design

Zhongzhiyuan, Beijing AI Origin and Dazhongsi carry environmental learning, knowledge compilation and public–service orchestration respectively. They share six public rights, four grounds and version governance while maintaining distinct plans, sections, users, rhythms and delivery conditions. [Depth](#)



Three key–area design index
Each key area is represented by an independent plan, key section and public–life scene.

5.1 Zhongzhiyuan AI Independent Innovation Acceleration Area

The primary chain links a potential station approach, public ground floors, park and Xiaoyue River; parallel streets and park paths provide backups. Public lobbies and forecourts punctuate the park edge. The riverbank supports ecological learning, exercise and community environmental issues. Low–speed devices and maintenance routes avoid peak pedestrian space. Environmental models disclose data and use fixed modes, physical shutdown and human takeover.

众智园 | 环境学习型开放地表

科研开放意图、蓝绿学习界面、横向接入与共享首层验证



QGIS verification: Zhongzhiyuan plan
QGIS verification: existing buildings in light grey; the station—park—river chain, public ground floors and ecological buffer on the real basemap.

Key-Area Detailed Design



Zhongzhiyuan plan

The station—park—river chain links potential transit access, public ground floors, environmental learning and the river, with separate backup paths and maintenance routes.



Zhongzhiyuan station—park—river section

Public ground floor, shared forecourt, primary walking/cycling route, maintenance route, ecological buffer and river learning unfold along one ground.



Zhongzhiyuan river—learning scene

Children, carers, wheelchair users, researchers and maintenance staff share shaded river paths and platforms outside the ecological buffer.

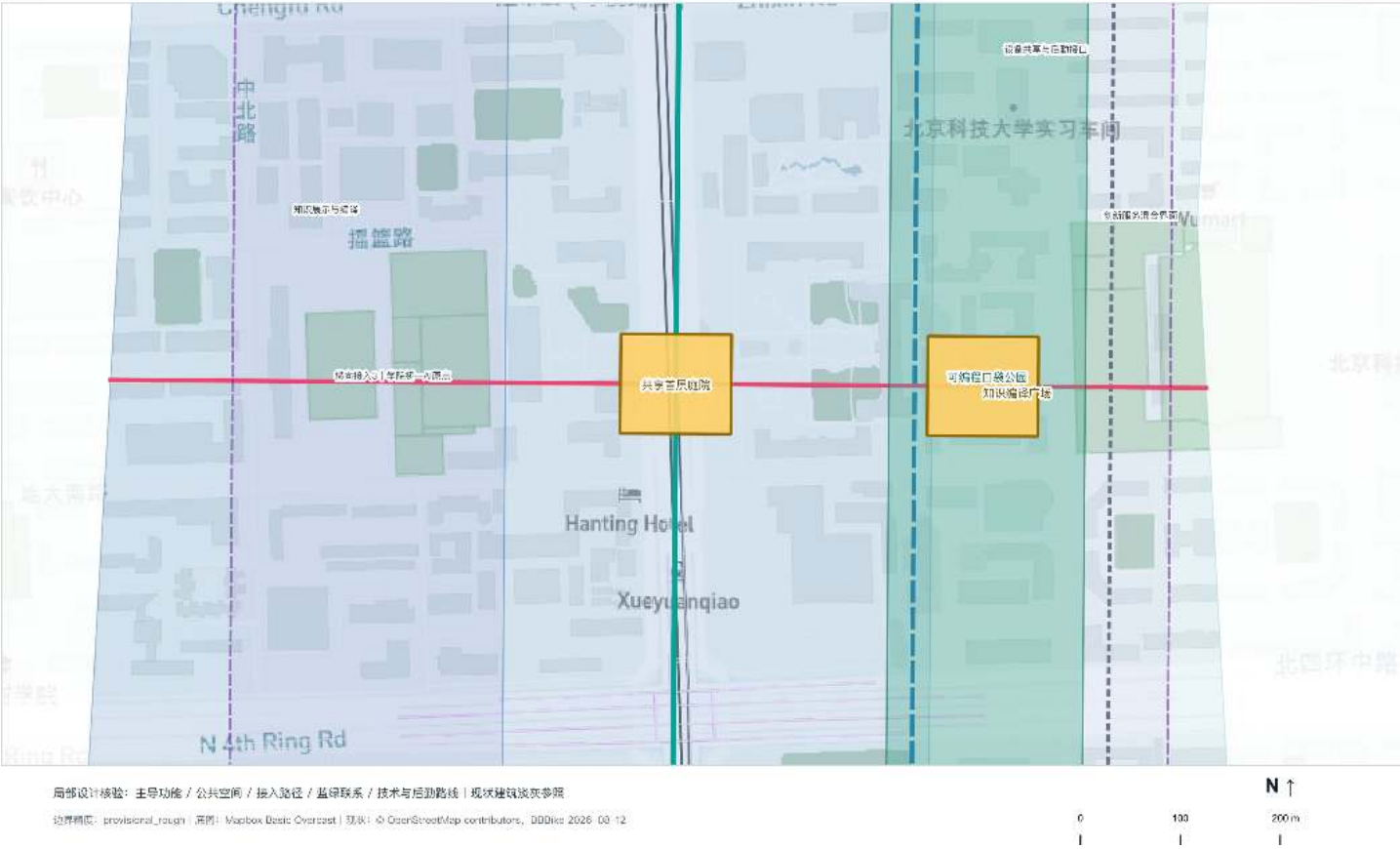
Key-Area Detailed Design

5.2 Beijing AI Origin Community

AI Origin combines Xueyuan Bridge access, an institutional public ground-floor cluster, shared forecourts and community paths. Every institution has a legible public entrance, an external evening route and independently operable ground floor. Urban model rooms, public AI plus human service, open courses, research interpretation, co-production and daily services occupy the public layer; controlled research retains separate security, logistics and fire safety.

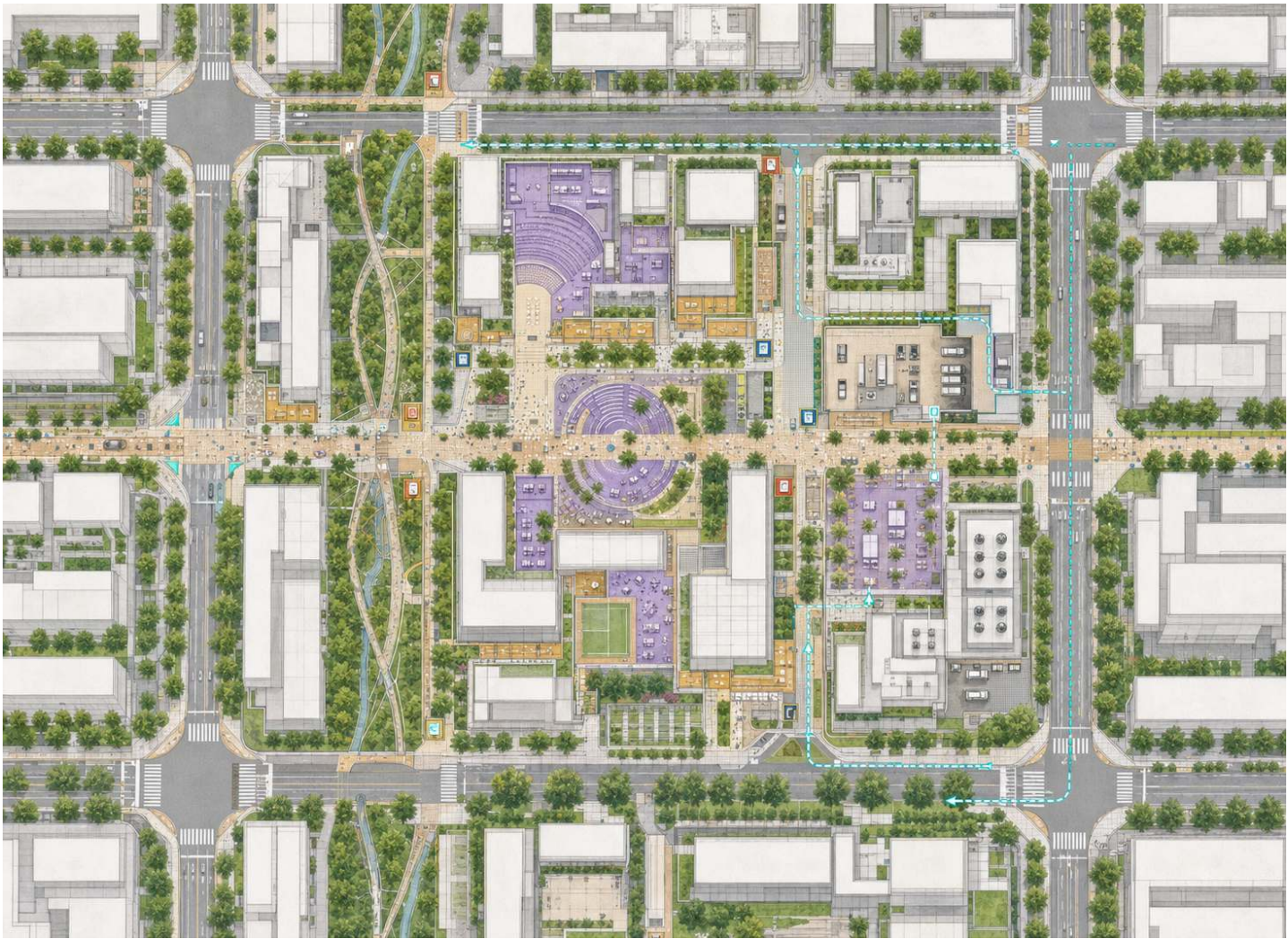
AI原点 | 知识编译型开放地表

知识编译广场、创新服务界面、步行接入与公共首层核验证



QGIS verification: AI Origin plan

QGIS verification: the knowledge street, institutional ground-floor cluster and logistics loop on the real basemap.



AI Origin plan

An east-west knowledge street, north-south evening routes, public forecourts and a logistics perimeter organise clear access gradients.



AI Origin public ground-floor section

Controlled research connects to open ground floors through separate security and logistics; the urban model room, public AI, human service, forecourt and street are zoned on the ground.

Key–Area Detailed Design



AI Origin evening knowledge scene
Independently operated public ground floors support model explanation, community consultation, open courses and everyday staying, with continuous human service and accessible seating.

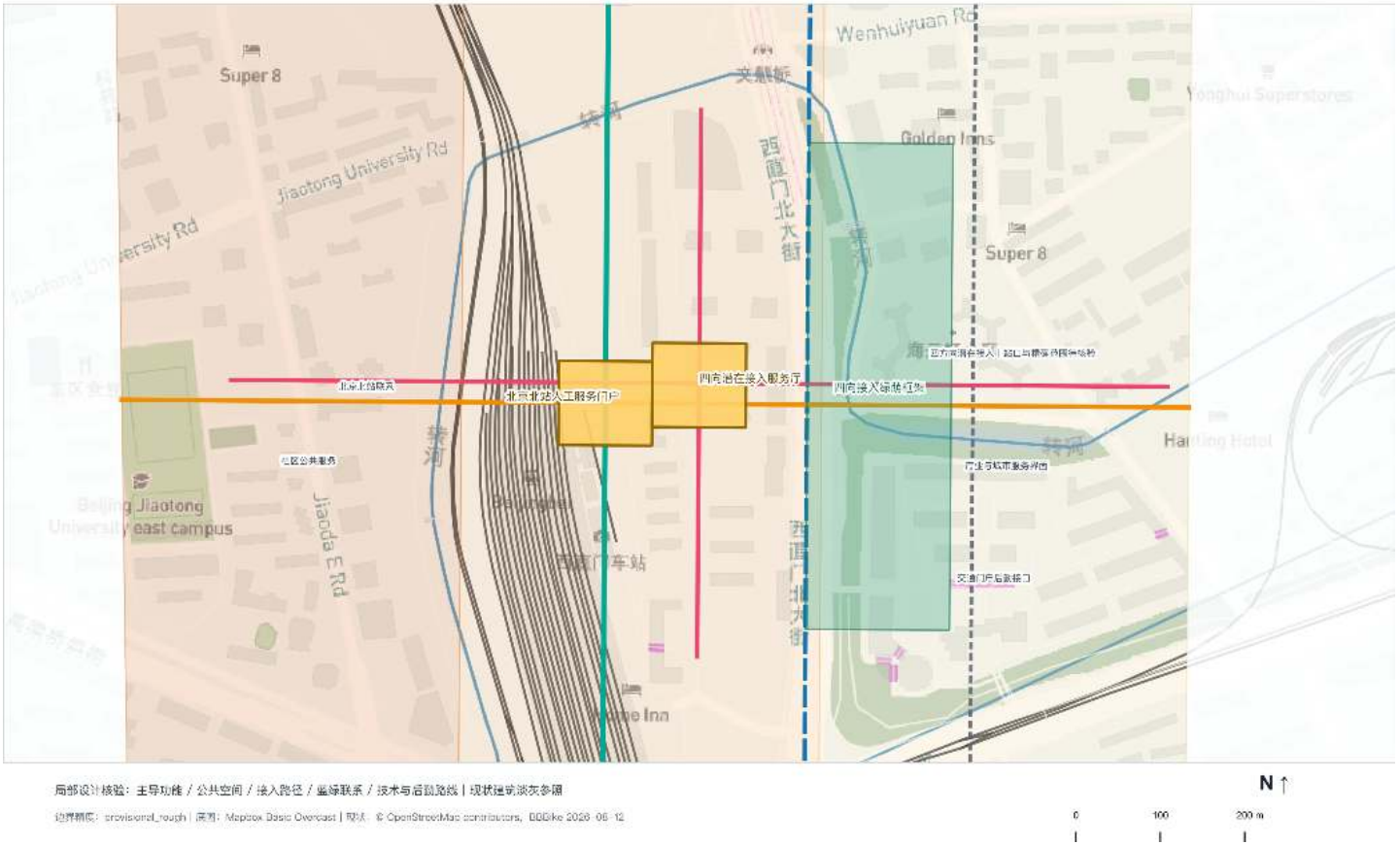
5.3 Dazhongsi AI Industry Cluster

Dazhongsi uses four potential access directions to organise continuous walking, cycling and accessibility along Zhichun Road—Dazhongsi—Jimen Bridge. Community and commercial ground floors combine employment, health, culture, learning, public AI, human service and basic daily needs. Crowd and capacity models assist wayfinding and resource scheduling; shared logistics and low-speed robots use time-separated routes.

The figure is a genuine two-dimensional plan focused on four potential directions, continuous cycling, community backup paths and rail-edge public space. PROV-KEY-003 lies near Beijing North Station and cannot be represented as the Dazhongsi official extent. Official extent, station exits, ownership, passenger flow, rail safety and crossing conditions remain unknown.

大钟寺框架 | 公共服务编排型开放地表

当前临时几何近北京北站；图面仅核临四方方向潜在接入框架



局部设计核编：主导功能 / 公共空间 / 接入路径 / 编修联系 / 技术与道路路线 | 现状建筑淡灰参照
边界精底：provisional_rough | 底图：Mapbox Basic Overcast | 图例：© OpenStreetMap contributors, © Esri 2025 05-12
QGIS verification: Dazhongsi four-direction access
QGIS verification: Dazhongsi shows only a four-direction potential-access framework without confirming station exits; redrawn after the official extent.

Key-Area Detailed Design



Dazhongsi two-dimensional plan

Building roofs, roads, water and railway form the reference. Dashed arrows mark potential access directions that require verification.



Dazhongsi street-community-rail-edge section

Community and commercial ground floors, service forecourt, road, safe crossing, walking/cycling, potential access forecourt and heritage green edge form the transverse sequence.



Dazhongsi high-flow public-service scene

Employment, health and community service, waiting, accessible crossing, cycling and rail-edge walking overlap at a busy street node.

Key–Area Detailed Design

5.4 Shared Standards, Different Outcomes

Key area	Public interface	Section relationship	Representative life	First package
Zhongzhiyuan	Station—ground floor—park—river	Research edge—public ground floor—forecourt—ecological buffer	Environmental courses, low-speed co-testing and river health	Primary/backup paths, public lobby, river classroom and maintenance interface
AI Origin	Xueyuan Bridge—knowledge street—ground-floor cluster	Controlled research—public ground floor—forecourt—street	Model deliberation, courses, co-production and community service	Evening entrance, urban model room, forecourt and logistics loop
Dazhongsi	Four potential directions—community/commercial ground floors—rail edge	Community ground floor—street—access forecourt—heritage green	Transfer help, employment, learning, health and culture	Cycling, accessible crossing, service forecourt and community entrance

6 AI Innovation Ecosystem, User Profiles and AI-Enabled Scenarios

Open-Source Ground organises a continuous loop: problem proposal → research and model → testing → public translation → adoption → feedback. Universities open problem statements, courses and shareable results; companies provide compliant testing and maintenance; communities and public bodies provide real problems and participate in evaluation; public space provides low-threshold experience, human service and exit channels. [Source](#)

6.1 Six User Groups

User	Daily need	Spatial and service response	Public safeguard
University researchers and teachers	Cross-institution collaboration, result explanation, real problems	Open courses, model forums, reserved workshops	Result licensing, professional responsibility, human review
Students, developers and start-up teams	Low-cost collaboration, testing, publishing and reputation	Evening public ground floors, shared test beds, tools and launch space	Public-benefit hours, training, exit and failure records
Company staff and enterprise-service workers	Commuting, display, recruitment, legal and supply-chain services	Station forecourts, public meeting rooms, employment and enterprise services	Stable space and hours for basic public service
Residents and carers	Access, health, education, maintenance and community activities	Fifteen-minute open units, human service, community forecourts	Offline channels and ordinary no-registration use
Children, older and disabled people	Accessibility, safety, rest, companionship and low cognitive load	Continuous accessibility, rest and hygiene, simple wayfinding, accompanied routes	Human and fixed facilities continue when technology stops
Visitors, commuters and international talent	Transfer, short stays, understanding the site and obtaining services	Multilingual wayfinding, railway culture, luggage help, public events	Personal data collected on a minimal basis

6.2 Twelve AI-Enabled Public-Life Scenarios

Scenario	Main location	Spatial configuration	Operation and safety
Low-speed embodied-system testing	Zhongzhiyuan campus-park buffer	Separate low-speed route, observation points, physical shutdown zone	On-site safety officer, time-slot operation, incident records
Adaptive blue-green maintenance	Xiaoyue River and parks	Environmental sensing, irrigation, shading and maintenance interfaces	Human maintenance first; fixed mode in flood and ecologically sensitive periods
Public urban-model deliberation	AI Origin public ground floor	Explainable-model table, source wall, discussion seating and human service	Published data sources, assumptions, versions and responsible persons
Public AI and human service	AI Origin knowledge street	Digital enquiry, human counter, private consultation and waiting	Human channel stays; corrections and appeals for results
Open-result compilation courses	AI Origin shared forecourt	Courses, result explanation, community topics and mobile facilities	Content reviewed by a responsible person; explicit knowledge licensing
Shared-robot last-metre handover	Institution edges and logistics loops	Low-speed route, handover point, storage/charging and maintenance	Separated from the accessible main route; human completes the final handover
Four-direction transfer assistance	Potential accesses around Dazhongsi	Fixed wayfinding, human service, rest and assistive equipment	Aggregated flows and active help requests
Employment and lifelong-learning lounge	Dazhongsi community public ground floor	Enterprise services, learning sessions, human advisers and community courses	Human review and explanation retained for recommendations
Health and culture daily station	Dazhongsi community forecourt	Health referral, cultural activity, rest and care support	Professional institution responsible; private information zoned
Jing-Zhang heritage knowledge route	Heritage park and portals along the corridor	History nodes, oral archives, courses and multilingual wayfinding	Facts, authors and copyright published with content
Accessible dynamic wayfinding	Five transverse access bands	Fixed wayfinding, facility status, human help and feedback points	Fixed system and human service continue together
Open urban-problem list	Each fifteen-minute open unit	Problem registration, project status, research matching and community discussion	Representative review, urgent referral, public closure and appeals

At least three serve as bounded testing and validation environments: low-speed embodied systems, public urban models and accessible wayfinding. Every scenario defines space, operator, limits, human review, shutdown and incident records.

6.3 Cultural Identity and Long-Term Appeal

Five functions structure the ecosystem: full-stack independent innovation, enterprise services, urban scenario access, talent/community service, and responsible governance. Three pilgrimage and honour-display landmarks — the Northern Environmental Learning Gate, the AI Origin Public Model Hall and the Southern Railway Knowledge Portal — connect railway history, Zhongguancun innovation and public AI ethics. A four-season programme publishes problems in spring, operates river learning in summer, hosts a Jing-Zhang Open Knowledge Week in autumn, and publishes model, fault, repair and exit records in winter.

7 Land Use, Building Scale and Demolish—Renovate—Retain Strategy

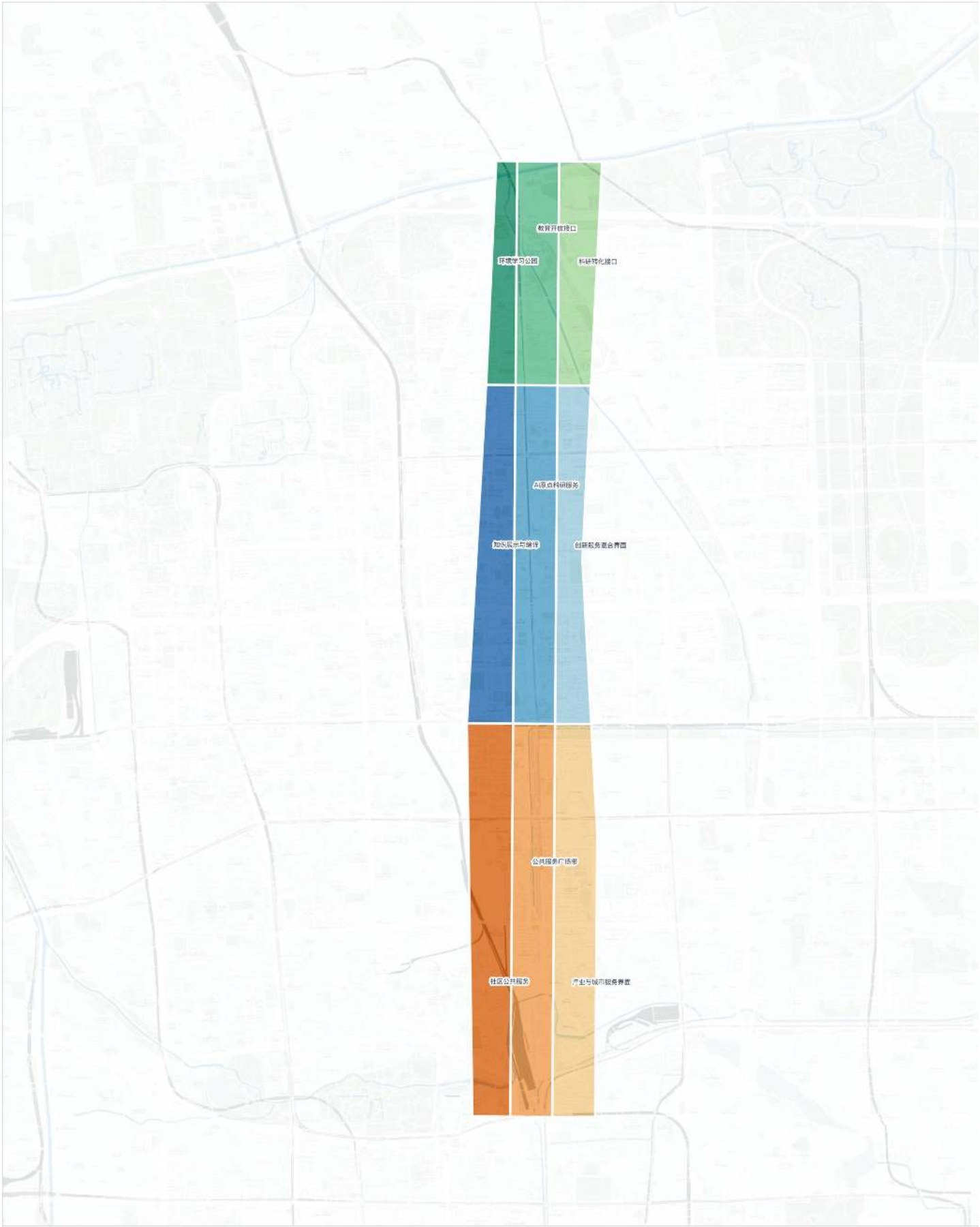
The land–use strategy follows “public ground first, mixed support, retained fabric and controlled insertion.” A continuous heritage–park and blue–green base supports public service, research and innovation, community life and mixed uses. Public ground floors and forecourts cut across conventional land categories as operational interfaces.

Land–use system	Overall layout	Design control
Railway heritage and park	A continuous north–south public base along the Jing–Zhang line	Retain railway engineering clues, public slow movement, open view corridors and cultural nodes
Blue–green ecology	Xiaoyue River, the railway green belt, campus and parkland, street rainwater space	Ecological buffer, flood elevation, shading, habitat and maintenance routes
Public service and open ground floors	Concentrated in the five transverse accesses, station forecourts and community entrances	Independent entrances, stable hours, basic service, accessibility and human staffing
Innovation and R&D	Maintain professional space for universities, research and enterprises	Connect to the city through public forecourts, courses, models and controlled facility interfaces
Mixed life and service	Mature communities in the south, central stations and institution edges	Combine employment, health, dining, learning, care and evening activity
Mobility and municipal	Stations, roads, cycling, logistics and equipment interfaces	Clear separation of primary/backup routes, time–slot logistics, fire and maintenance
Controlled areas awaiting verification	Boundaries, station exits, crossings, river and ownership conditions pending	Retain multiple options and reversible projects; precise control after materials arrive

The current building evidence identifies 120 OSM–derived footprints only as retain/adapt/public–ground–floor opportunities. It does not assign demolition. Ownership, footprint survey, structure, fire safety, height, condition, heritage value and use require building–by–building verification. FAR, building height, BCR, road red lines and legal development capacity remain unknown.

专题01 | 功能与规划分区

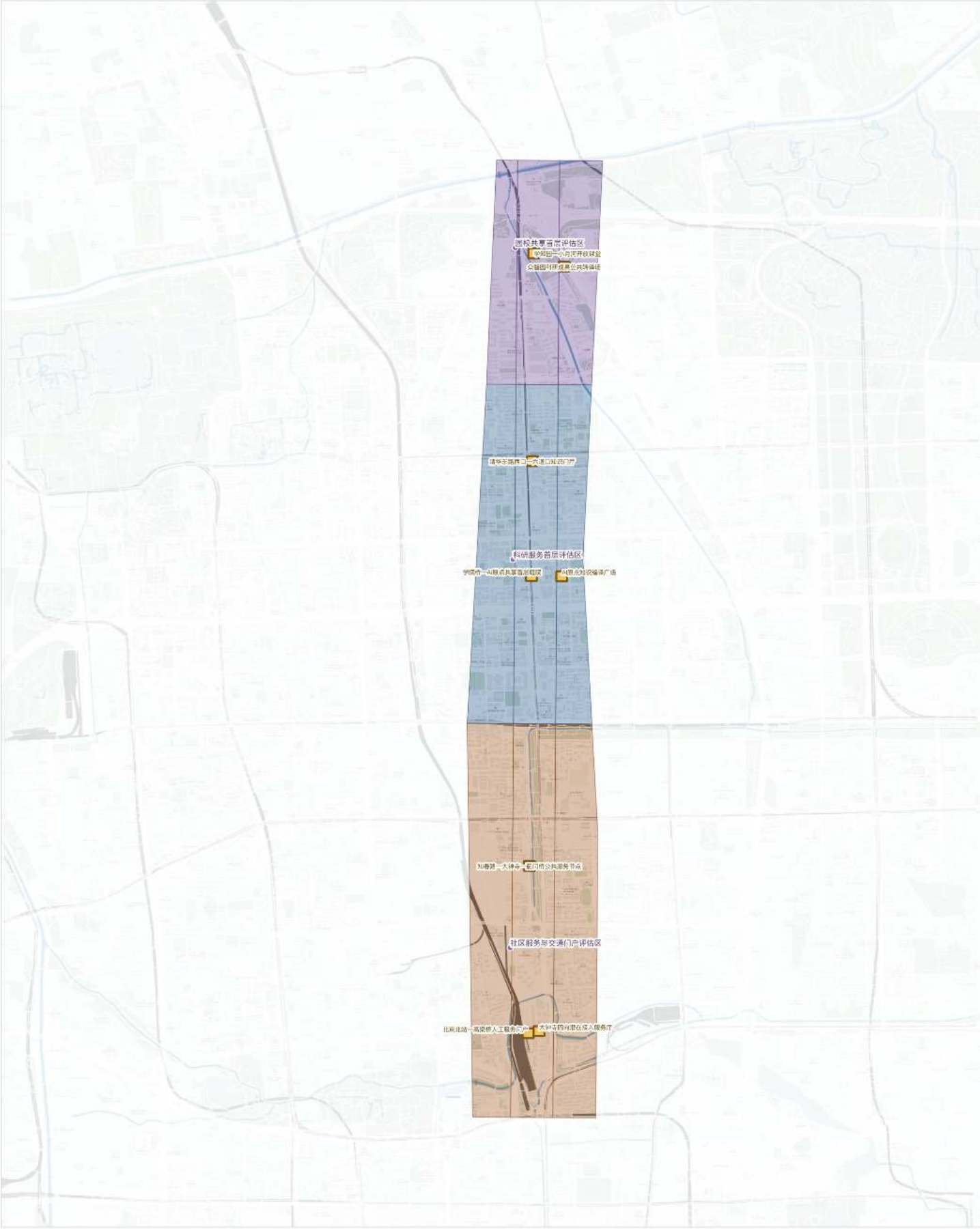
九个概念主导功能分区 | 北段环境学习、中段知识编译、南段公共服务



九个细分区及主导功能：法定用地性质与指标由控规核验

专题04 | 公共首层与建筑更新策略

三类更新评估区与公共首层接口节点 | 逐栋机会对象保留于工程核验层



Building category	Judgement basis	Design action	Next-step material
Retain in use	Stable use, suitable structure, clear cultural or community value	Improve entrance, accessibility, wayfinding and external public space	Ownership, current use, fire safety and heritage value
Adaptive renovation	Convertible space, location connects to the public ground	Insert courses, health, tools, display, community and enterprise services	Structure, equipment, fire, security and operator
Public ground-floor conversion	At the intersection of access bands and public space	Independent entrance, shared forecourt, evening hours and logistics separation	Ground-floor area, thresholds, opening protocol and public-benefit quota
Awaiting professional evaluation	Building quality, efficiency or engineering conditions not yet surveyed	Retain multiple retain/renovate/remove paths	Survey, appraisal, ownership, control plan and economic assessment

8 Mobility, Transit, Municipal and Public–Service Systems

The mobility hierarchy gives priority to continuous walking and accessibility, then cycling, transit connection, shared service, logistics and essential motor access. Five transverse accesses combine crossings, shade, seating, fixed and dynamic wayfinding, cycle parking/repair and potential station links. Exact exits, rail–crossing structures, road red lines and traffic organisation await transport and rail review.

Journey chain	Primary–path requirement	Backup and conflict control	Verification conditions
Station—public ground floor	Shortest legible route linking station service, accessible crossing and public entrance	Peak flow diversion, visible evening route, human wayfinding	Station exits, opening hours, flows, gradients, crossing waits
Community—heritage park	Multi–entrance walking and cycling access	Parallel street backup; staggered school and residential entrances	Community entrances, ownership, parking, night safety
Campus—riverbank	Campus public lobby links safe riverbank slow movement and stopping points	Maintenance route separated from public path; high–level route in floods	River, flood, ecological buffer, campus opening
Delivery—equipment interface	Low–speed time–slot access at building rear or forecourt edge	Paused during peak and events; separated from accessible main chain	Loading capacity, fire, noise, insurance and responsibility chain

The municipal framework places sensing, communications, energy, water, storage, charging, repair and emergency shutdown in maintainable interfaces. Equipment remains replaceable and physically stoppable. Public toilets, drinking water, rest, first aid, human information and universal access are treated as basic infrastructure. Ownership, capacity, flood, fire, power, drainage and network–security conditions remain unknown.

专题02 | 交通与开放接入

连接公共地表、平行备选线、五条横向街道、潜在站点接入与慢行网络

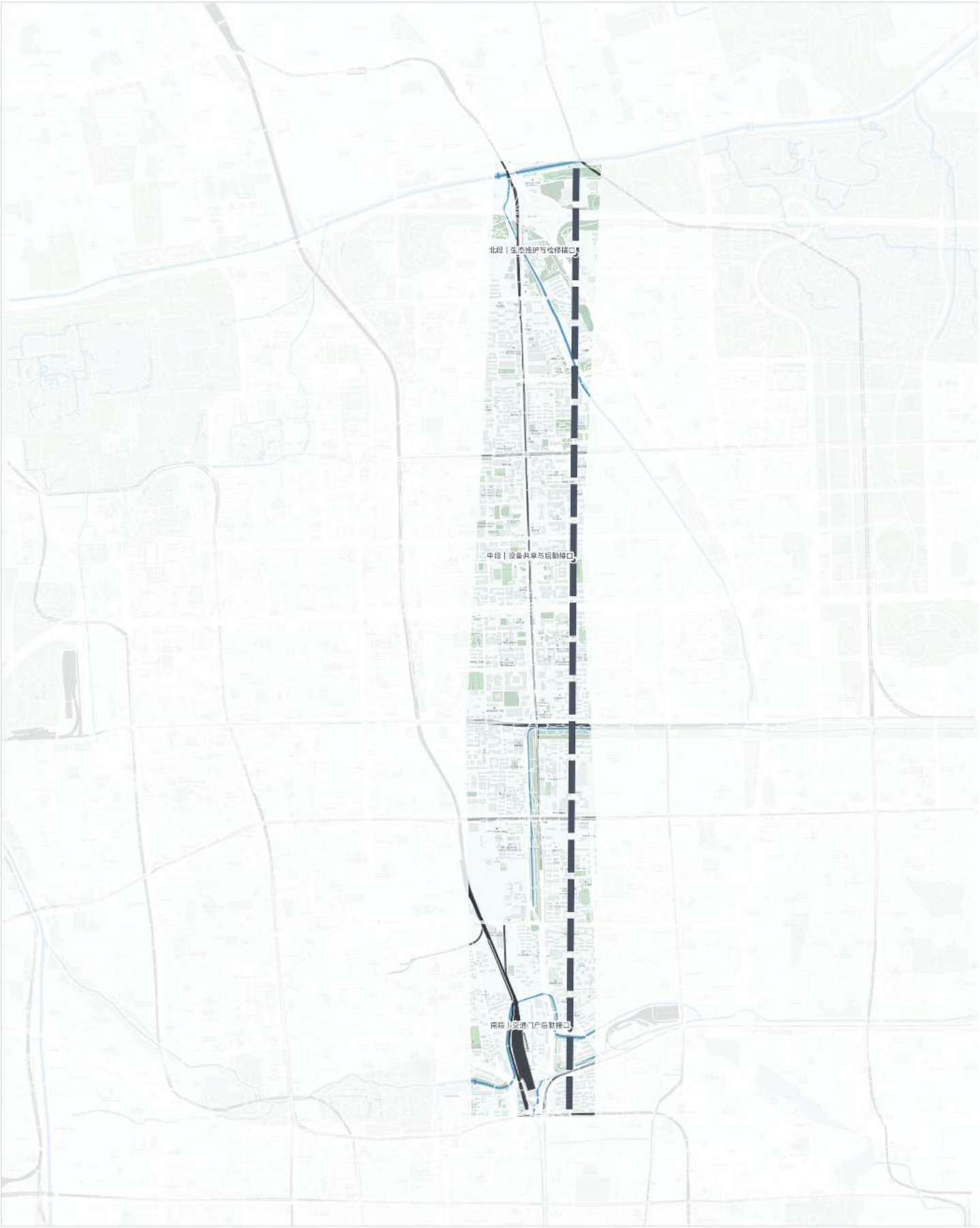


连接主脊 / 平行备选线 / 横向接入1-5 / 大佛寺回南道在步行接入



专题05 | 技术设施与后勤

独立技术设施服务走廊及现状交通、市政空间参照



技术设施服务走廊：市政管量、检修界面和权属条件由专项核验

04 交通与蓝绿共同构成可达地表

五条横向接入同时处理站点、过街、骑行、无障碍、首层和停留



Mobility and blue-green system

Five transverse accesses make mobility, public ground floors, blue-green space and service a single project package.

9 Blue–Green Space, Public Space and Urban Character

The blue–green system links Xiaoyue River, the heritage park, institution and street green space, community forecourts, rain gardens and pocket spaces. The north prioritises ecological learning and flood–adaptive operation; the centre prioritises shade, campus–edge continuity and outdoor learning; the south prioritises tree–lined streets, service forecourts and rail–edge ecological links.

Blue–green object	Spatial outcome	Knowledge and public use	Engineering and maintenance conditions
Xiaoyue River and banks	Ecological buffer stays continuous; slow movement at safe elevation; course platforms avoid sensitive areas	Environmental monitoring explanation, seasonal courses, community health, problem registration	River extent, flood elevation, habitat, water safety and maintenance
Railway heritage park	Continuous tree–lined path links transverse streets; railway clues and open view corridors retained	Railway engineering history, urban expansion, daily sport and rest	Heritage alignment, management boundary, vegetation and facilities
Streets and shared forecourts	Rain gardens, shading, seating and safe crossings organise staying	Thermal–comfort explanation, community activity, institutional spillover	Underground utilities, sight distance, drainage, fire and accessibility
Roofs and courtyard edges	Store, drain, cool and supplement habitat	Institutional courses, maintenance training and long–term monitoring	Structure, drainage, fire, security and upkeep funding

Public space ranges from continuous base and transverse access to shared forecourt, public ground floor and micro–facility. A common railway material and linear order, transparent ground floors, restrained lighting, integrated roof equipment and legible public/controlled edges establish shared character. Chinese, English, simple graphics and non–visual cues communicate historical facts, current status, responsibility and knowledge sources.

专题03 | 蓝绿与公共空间

河岸、公园绿地、环境学习带、蓝绿缝合与八个公共节点



10 Renewal Projects, Policy and Phased Implementation

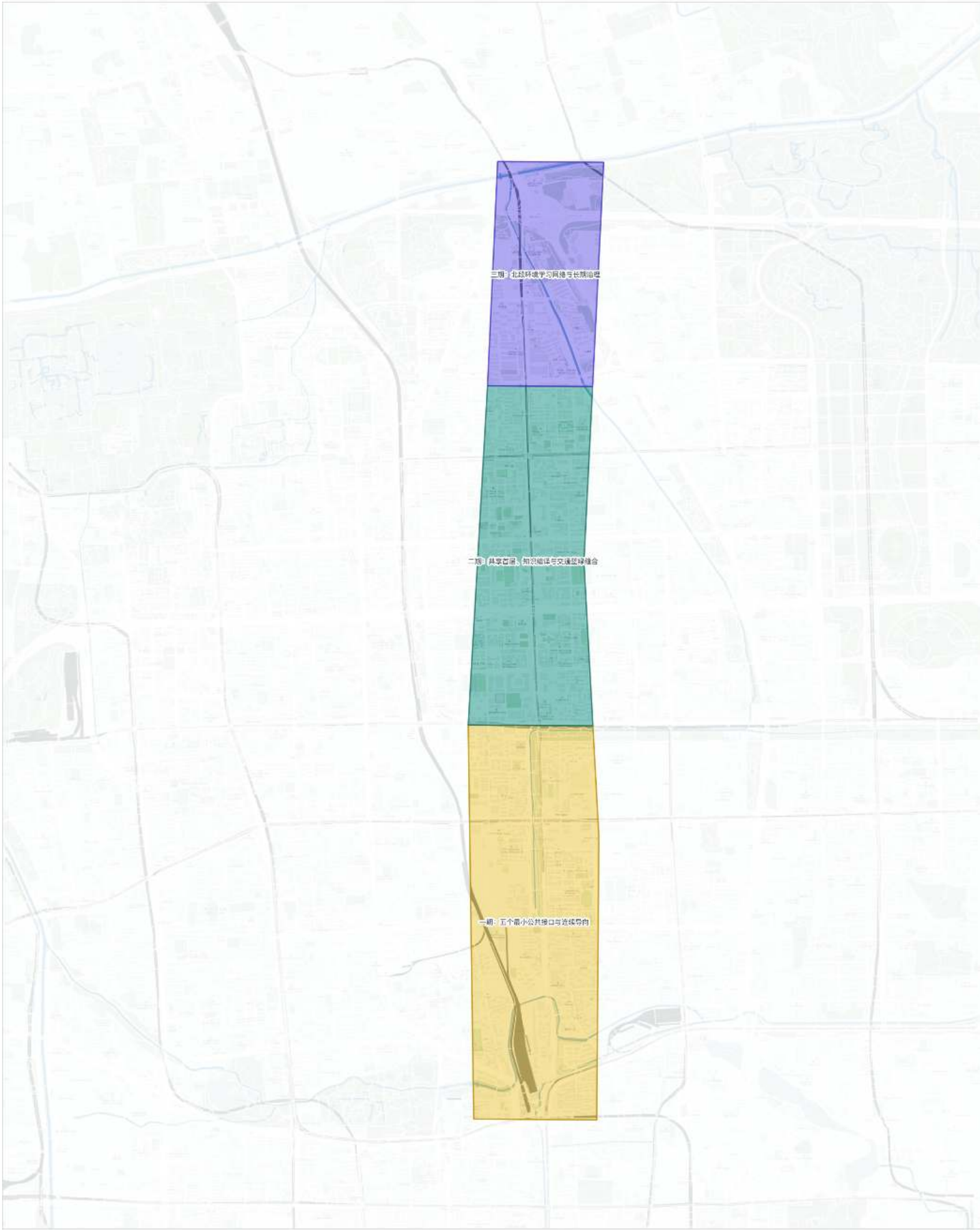
Near-term projects establish minimum viable ground: accessibility, wayfinding, first public ground floors, river courses, human service and open ledgers. Growth-stage projects deliver five accesses, Fifteen-Minute Open Units, complete key-area packages, shared equipment and public models. Long-term work expands the heritage knowledge route, corridor governance, evaluation and structurally dependent crossings. [Spatial data](#)

Phase	Project package	Direct outcome	Dependency
Near term: minimum viable ground	Three priority key spatial units, basic wayfinding and accessibility, first public ground floors, riverbank courses, human service and open ledgers	A reachable, usable, feedback-ready prototype	Entrances, hours, fire, security, river and operator
Growth: multi-unit collaboration	Five transverse accesses, fifteen-minute open units, full key-area packages, shared equipment and public models	A cross-institution service and scenario network	Multi-party protocols, maintenance funding, data and professional review
Long term: continuous renewal	Corridor-wide versioning, heritage knowledge route, indicator post-review, mature project expansion and low-performing project exit	A continuously learning public ground	Long-term operator, public oversight, annual audit and statutory linkage

Each Ground Protocol Unit combines spatial-access, data-model and operations-maintenance agreements. Governance brings together public coordination, professional review, institutional operators, community representatives and independent evaluation. AI supports problem classification, option generation, operational prediction and evaluation; accountable professional and public bodies decide spatial trade-offs, public resources and rights impacts.

专题06 | 实施分期

一期最小公共接口、二期知识与蓝绿缝合、三期环境学习与长期治理



一期 0—12个月 / 二期 1—3年 / 三期 3—6年

N ↑

11 Metrics, Area Recalculation and Compliance

Metrics are grouped as existing evidence, design supply and operational performance. Geometry recalculation gives a provisional Overall Design Area of approximately 11.413 km² and three provisional key–area extents of approximately 1.929 km², 1.043 km² and 0.720 km². These numbers are low–confidence geometry derivatives, not official control figures. [Depth](#)

Indicator group	Current baseline or direction	Design meaning	Review method
Scope and levels	About 43.6 km ² research, 11.4 km ² overall design, three key areas	Control the three working scales and deliverable depth	Recalculate in EPSG:4548 when official extents arrive
Transit and life circles	Median station distance ~606 m; 10–min walk ~48.2%, 15–min ~78.6%, 5–min cycle ~85.5%	Set the priority of transverse accesses and real entrances	Field walking, entrances, waits, gradients and hours
Blue–green and buildings	Research grid green ~13.2%, water ~0.7%, building cover ~17.8%	Support three–section differences, shading, blue–green links and ground–floor selection	Recalculate with official land use, survey and river materials
Key–area differences	Zhongzhiyuan stronger blue–green; AI Origin concentrated innovation services; Dazhongsi high–flow public service	Determine environmental–learning, knowledge–compilation and service–orchestration prototypes	Recalculate when official key–area extents and parcels arrive
Public–ground supply	Public entrances, accessible chains, public ground floors, outdoor space, equipment interfaces, human service	Test the minimum usable state of each open unit	Survey and accept unit by unit
Operational performance	Opening hours, public–benefit quota, use, faults, complaints, repair, exit	Judge public benefit and maintenance capacity	Annual public reporting per unit after operation

Core public–benefit measures are: more open entrances and stable hours; shorter real walking and accessible chains; broader public ground–floor and basic–service coverage; better blue–green continuity and thermal comfort; more urban problems entering research and service; and queryable fault, complaint, repair and exit processes. Every measure keeps source, time, extent, method and status.

05 指标沿同一证据链复算

现状基线解释选点，设计供给检验最低公共能力，运营绩效判断持续收益



SOURCE → ANALYSIS → GEOMETRY → DESIGN → OPERATION → REVIEW

几何复算：总体临时范围约11.413 km²；临时性、低置信度与未知法定指标均在metrics、sources和assumptions中记录。

Metrics and evidence recalculation
Existing baselines, design supply and operational performance connect analysis, spatial objects and annual evaluation along one evidence chain.

12 Risk, Copyright and Compliance

The submission is a conceptual urban–design and implementation framework. Its precision supports spatial structure, priority, key–area prototypes and delivery conditions. Official boundaries, statutory plans, station/road data, ownership, building survey, river/flood information, municipal capacity, investment and rail–crossing engineering are required before parcel, engineering or construction decisions. [Depth](#)

Key condition	Current expression	Follow–up work
Provisional boundary and Dazhongsi extent	Overall structure uses provisional research extent; Dazhongsi shows four–direction potential access	Redraw plans and recalculate after official extent arrives
Station exits, ownership and crossings	Drawings retain potential access directions and ground improvement	Verify rail, road, railway, ownership, clearance and construction
Ground floors, fire and security	Expressed as opening type, entrance and function mix	Building–by–building survey, appraisal, fire/security review and negotiation
River, stormwater and ecology	Ecological buffer, high–level backup and seasonal operation	Water, flood, habitat, water–safety and maintenance studies
Data and model rights	Minimisation, aggregation, tiered access, human review, deletion and appeal	Dedicated review for personal, medical, traffic–safety and professional–decision scenarios
Technology dependency and maintenance	Basic public service runs independently; equipment has shutdown, replacement and recovery	Define maintainer, funding, spares and fault response
Public–commercial balance	Public–benefit hours, basic service and public return enter the open protocol	Annual review of hours, accessibility, basic service and fairness

The Overall Design Area and all three key areas retain `provisional_constraint`, `official_boundary=false` and `boundary_precision=provisional_rough`. Dazhongsi keeps four potential access directions without claiming a station entrance or precise extent. Unknown statutory land–use controls, FAR, building heights, road red lines, station exits, crossings, river and municipal conditions remain `unknown` in the structured evidence.

Case images trace directly to authors, publishing institutions, original pages and open licences. QGIS analysis uses OpenStreetMap data under ODbL, with retrieval date and transformations recorded. Project–team concept images are spatially constrained by QGIS basemaps and formal GeoJSON. Existing estimates support comparison and selection; case claims use original official, public–institution, university, audit or peer–reviewed sources; spatial, policy and operating actions remain conceptual recommendations for professional development.

13 References

1. Beijing Municipal Commission of Planning and Natural Resources, Haidian Branch, *Qualification Announcement for the Centennial Jing–Zhang AI Innovation Belt International Urban Design Call*, 2026. [Source](#) 2. *Centennial Jing–Zhang AI Innovation Belt Agent Taskbook*, Formal Submission Guide and public site materials. [Source](#) 3. OpenStreetMap contributors; BBBike Beijing extract, file dated 8 August and retrieved 12 August 2026; Open Database License 1.0. [Source](#) 4. King’s Cross development operator; Camden Council; Greater London Authority, *Opportunity Area Portrait: King’s Cross*, 2025. 5. SEMAPA, *Paris Rive Gauche*; Cour des comptes, SEMAPA/Paris Rive Gauche audit report, 2012–2019. 6. Shanghai Yangpu District People’s Government; Wu, Y. and Liu, Y., “Transforming Industrial Waterfronts into Inclusive Landscapes,” *Sustainability* 15(6):5060, 2023. 7. Shui On Xintiandi, KIC Shanghai; Shanghai Yangpu District People’s Government, KIC twenty–year review. 8. Barcelona City Council, *22@ Barcelona Innovation District 2024*; Barcelona City Council and Barcelona Activa, *Impulsem el 22@*, 2020. 9. MIT, *Kendall Square Initiative* and *Planning the Transformation of Kendall Square*; Dávila, J., *Spatial Qualities of Innovation Districts*, MIT DUSP, 2013. 10. Seoul Metropolitan Government, *Gyeongui Line Forest Park*; The Bentway, *About*; PUBLIC WORK, *The Bentway*. 11. City of Stockholm, *Stockholm Royal Seaport*; C40, *Stockholm Royal Seaport Good Practice*; Shenzhen Urban Management authority, *Guiwan Park*. 12. Beijing SASAC, Shougang transformation; Hong Kong Science and Technology Parks Corporation; Hong Kong Planning Department, Science Park phasing and land use.